

Eleven Primitives and Three Gates: The Universal Structure of Computational Imaging

Chengshuai Yang^{1,*} Xin Yuan²

Abstract

Computational imaging systems—from coded-aperture cameras to cryo-electron microscopes—span five carrier families yet share a hidden structural simplicity. We prove that every imaging forward model decomposes into a directed acyclic graph over exactly 11 physically typed primitives (Finite Primitive Basis Theorem)—a sufficient and minimal basis that provides a compositional language for designing any imaging modality¹. We further prove that every reconstruction failure has exactly three independent root causes: information deficiency, carrier noise, and operator mismatch (Triad Decomposition). The three gates map to the system lifecycle: Gates 1 and 2 guide design (sampling geometry, carrier selection); Gate 3 governs deployment-stage calibration and drift correction. Validation across 12 modalities and all five carrier families confirms both results, with +0.8 to +13.9 dB recovery on deployed instruments. Together, the 11 primitives and 3 gates establish the first universal grammar for designing, diagnosing, and correcting computational imaging systems.

Introduction

Every imaging instrument—whether it captures spectra, spins, X-rays, or sound waves—converts a physical scene into digital measurements through a chain of transformations: a wave propagates, interacts with the object, passes through optics, and is recorded by a detector. Computational imaging reconstructs the scene by inverting this chain. The reconstruction algorithm assumes a mathematical model of the chain (the “forward model”), but this model inevitably diverges from reality: optics shift, detectors drift, and calibration degrades. When the model is wrong, the reconstruction fails—not because the algorithm is weak, but because it is solving the wrong problem.

This paper establishes that the space of imaging forward models has a surprisingly simple and universal structure—a finite grammar of 11 primitives—and that every reconstruction failure admits a complete, actionable diagnosis through exactly 3 independent gates. Over the past decade, the community has invested enormous effort in improving

¹NextGen PlatformAI C Corp, USA. ²School of Engineering, Westlake University, Hangzhou, China. *Correspondence: integrityyang@gmail.com

reconstruction algorithms—progressing from compressed sensing^{2,3} and plug-and-play priors^{4,5} to deep unrolling networks⁶ and vision transformers⁷—with snapshot compressive imaging emerging as a unifying framework for coded aperture systems⁸. These advances are real and important: a better solver on a well-calibrated operator remains the gold standard. Yet algorithms routinely fail when deployed on real instruments⁹, and the root cause is often not the algorithm but an undiagnosed forward-model error. Calibration methods exist for specific instruments—ESPIRiT¹⁰ for MRI, ePIE¹¹ for ptychography—but they do not generalise across modalities, and no systematic framework decomposes reconstruction failures into root causes or guides the design of new imaging systems from first principles.

Here we establish two complementary foundations within Physics World Models (PWM). The **Finite Primitive Basis Theorem** (Theorem 3) proves that every imaging forward model in a broad operator class $\mathcal{C}_{\text{img}}$ admits an ε -approximate representation as a typed directed acyclic graph (DAG) over exactly 11 canonical primitives—and that this library is minimal¹. The **Triad Decomposition** proves that every reconstruction failure decomposes into three gates—information deficiency (**Gate 1**), carrier noise (**Gate 2**), and operator mismatch (**Gate 3**)—each independently testable and independently addressable. The 11 primitives *enable* the 3-gate diagnosis: the DAG structure localizes each gate to specific physical operators, making both design and correction actionable.

The framework serves dual purposes. For **existing instruments**, the Triad identifies the dominant gate and prescribes targeted intervention—calibration for Gate 3, acquisition redesign for Gate 1, or source/detector improvement for Gate 2. For **new instruments**, the 11-primitive basis provides a compositional design language: specifying a new modality reduces to selecting primitives and their parameters, with Gate 1 analysis guiding sampling strategy, Gate 2 guiding carrier and source selection, and Gate 3 predicting the calibration accuracy required for a target reconstruction quality. Solver design remains essential throughout: once the dominant gate is addressed, a better solver delivers further gains on the corrected operator.

We validate both results across twelve modalities spanning all five carrier families—optical photons (CASSI¹², CACTI¹³, SPC¹⁴, lensless, compressive holography, fluorescence microscopy), X-ray photons (CT¹⁵, cone-beam CT), electrons (ptychography¹⁶, cryo-EM), nuclear spins (MRI^{17,18}), and acoustic waves (ultrasound)—with hardware validation on all twelve modalities confirming Triad predictions on physical measurements. The validated modalities include two Nobel Prize-winning techniques (cryo-EM, 2017 Chemistry; super-resolution fluorescence, 2014 Chemistry), three clinically deployed instruments (CT, CBCT, MRI), and the first acoustic-carrier validation in computational imaging. A held-out closure test on 8 additional modalities confirms basis completeness with saturating growth. The proof of Theorem 3 is in Supplementary Note 12; formal primitive semantics and typed DAG denotation are developed in¹.

Why this matters beyond computational imaging. The finding that eleven primitives span imaging forward models raises a question for future work: do similar finite-basis results hold for other measurement domains? The OPERATORGRAPH formalism and Triad Decomposition are structurally applicable wherever a forward model maps quantities of interest to observations through composable physical stages. Seismology (wave propagation + attenuation + sensor response), radio astronomy (aperture synthesis + ionospheric distortion + noise), and particle physics (interaction model + detector response + pile-up) all exhibit the same three-gate failure structure. Whether these domains also admit small primitive bases is an empirical question that our framework could help answer.

The Finite Primitive Basis

A foundational question in computational imaging is whether the diversity of imaging forward models—from coded aperture cameras to MRI scanners to electron microscopes—can be captured by a small, fixed set of primitive operators. We answer affirmatively.

The OperatorGraph representation. Every imaging forward model is encoded as a typed directed acyclic graph (DAG) in which each node wraps a single primitive physical operator and edges define the data flow from source to detector. Every primitive implements both a `forward()` and an `adjoint()` method, with validated adjoint consistency ensuring $\langle H\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, H^\dagger \mathbf{y} \rangle$ to within numerical precision. Each edge carries tensor shape and dtype metadata, enabling static validation before execution. We call this formalism the OPERATORGRAPH intermediate representation (IR).

Eleven canonical primitives. The OPERATORGRAPH IR defines a library of exactly 11 canonical primitives $\mathcal{B} = \{P, M, \Pi, F, C, \Sigma, D, S, W, R, \Lambda\}$:

#	Primitive	Notation	Physical action
1	Propagate	$P(d, \lambda)$	Free-space wave propagation
2	Modulate	$M(\mathbf{m})$	Element-wise multiplication (mask, coil, absorption)
3	Project	$\Pi(\theta)$	Radon line-integral projection
4	Encode	$F(\mathbf{k})$	Fourier-domain encoding (k -space)
5	Convolve	$C(\mathbf{h})$	Spatial convolution (PSF)
6	Accumulate	Σ	Summation over spectral/temporal axis
7	Detect	$D(g, \eta)$	Detector response (5 canonical families)
8	Sample	$S(\Omega)$	Sub-sampling on index set Ω
9	Disperse	$W(\alpha, a)$	Wavelength-dependent spatial shift
10	Scatter	$R(\sigma, \Delta\varepsilon)$	Direction change and/or energy shift
11	Transform	$\Lambda(f, \theta)$	Pointwise nonlinear physics operation

The Detect nonlinearity η is restricted to five canonical families (linear-field: $\eta(x) = gx$; logarithmic; sigmoid; intensity-square-law: $\eta(x) = g|x|^2$; coherent-field), each with at most 2 scalar parameters. The Transform primitive Λ applies a pointwise nonlinear function

94 within the physics chain (not at the detector) and is restricted to five families: exponential
 95 attenuation ($e^{-\alpha x}$, Beer–Lambert), logarithmic compression, phase wrapping ($\arg(e^{ix})$),
 96 polynomial ($\sum a_k x^k$, $d \leq 5$), and saturation. Both Detect and Transform are pointwise
 97 and parameter-limited, preventing either from becoming a universal approximator (see¹ for
 98 formal definitions and non-universality proofs).

99 **Physics-stage mapping.** Each factor of a forward model is classified into one of six
 100 physics-stage families and mapped to primitives from $\mathcal{B}$: propagation $\rightarrow \{P, C\}$; elastic in-
 101 teraction $\rightarrow \{M\}$; inelastic interaction (scattering) $\rightarrow \{R\}$; pointwise nonlinear physics $\rightarrow$
 102 $\{\Lambda\}$; encoding–projection $\rightarrow \{\Pi, F\}$; detection–readout $\rightarrow \{\Sigma, S, W, C, D\}$. This classifica-
 103 tion formalizes the source–medium–sensor decomposition that structures classical imaging
 104 theory¹⁹.

105 **Fidelity levels.** Forward models can be written at increasing physical fidelity: linear
 106 shift-invariant (simplest), linear shift-variant, nonlinear ray/wave-based, and full-wave or
 107 Monte Carlo. All 11 primitives apply uniformly across every fidelity level. Linear stages are
 108 composed from the first 10 primitives; nonlinear stages (beam hardening, phase wrapping,
 109 saturation) are captured by Transform Λ . The theorem therefore covers the full fidelity
 110 spectrum without restriction.

111 **Physical carriers.** The template library spans five carrier families—optical photons, X-
 112 ray photons, electrons, nuclear spins, and acoustic waves—with particle-beam probes (neu-
 113 trons, protons, muons) representable in the extended registry using *zero new primitives*:
 114 neutron CT reuses the X-ray CT DAG ($P \rightarrow \Lambda \rightarrow D$) with nuclear cross-section param-
 115 eters; proton CT inserts a Bethe–Bloch stopping-power stage via Transform Λ (#11); muon
 116 tomography uses Scatter R (#10, already introduced for Compton imaging) for multiple
 117 Coulomb scattering and POCA reconstruction (Supplementary Note S24). In total, 170
 118 modality templates cover 23 categories (microscopy, medical imaging, electron microscopy,
 119 remote sensing, spectroscopy, and others; see¹), of which 12 now have full end-to-end cor-
 120 rection validation across all five primary carrier families (Supplementary Table S3).

121 **Theorem statement.**

122 **Definition 1** (Imaging Operator Class). $\mathcal{C}_{\text{img}}$ consists of all imaging forward models ex-
 123 pressible as a finite sequential-parallel composition of stages—each either linear with bounded
 124 operator norm ($\|H_k\| \leq B$) or pointwise nonlinear with bounded Lipschitz constant ($\text{Lip}(\Lambda_k) \leq$
 125 L)—with stage count $K \leq N_{\text{max}}$.

126 **Definition 2** (ε -Approximate Representation). A typed DAG G with nodes $V \subseteq \mathcal{B}$ is an
 127 ε -approximate representation of $H \in \mathcal{C}_{\text{img}}$ if $\sup_{\|\mathbf{x}\| \leq 1} \|H(\mathbf{x}) - H_G(\mathbf{x})\| / (\|H(\mathbf{x})\| + \delta) \leq \varepsilon$,
 128 where $\delta > 0$ is a regularization constant, and $|V| \leq N_{\text{max}}$, $\text{depth}(G) \leq D_{\text{max}}$.

129 **Theorem 3** (Finite Primitive Basis). *For every $H \in \mathcal{C}_{\text{img}}$, there exists a typed DAG G with*
 130 *$V \subseteq \mathcal{B}$ that is an ε -approximate representation of H .*

131 *Proof sketch.* By Definition 1, $H = H_K \circ \dots \circ H_1$ with $K \leq N_{\text{max}}$. Each factor is classified
 132 into one of six physics-stage families and realized by primitives: propagation $\rightarrow P$ or C ; elas-
 133 tic interaction $\rightarrow M$; inelastic interaction $\rightarrow R$; pointwise nonlinear physics $\rightarrow \Lambda$; encoding-
 134 projection $\rightarrow \Pi$ or F ; detection-readout $\rightarrow$ finite composition from $\{\Sigma, S, W, C, D\}$. For
 135 linear stages, per-factor errors compose via sub-multiplicativity: $\|H - H_G\| \leq K \cdot \max_k(\varepsilon_k) \cdot$
 136 B^{K-1} . For nonlinear stages, the Lipschitz composition bound $\|H(\mathbf{x}) - \hat{H}(\mathbf{x})\| \leq \sum_k \varepsilon_k \prod_{j>k} \gamma_j$
 137 applies, where γ_j is the Lipschitz constant of stage j . Both bounds yield $\leq \varepsilon$ under the
 138 stated regularity conditions. Full proof in Supplementary Note 12; formal primitive seman-
 139 tics and typed DAG denotation in¹. Example OPERATORGRAPH DAGs for CASSI, MRI,
 140 and CT alongside the four-tier fidelity ladder are shown in Figure 2. $\square$

141 **Scope of $\mathcal{C}_{\text{img}}$.** Covers all clinical, scientific, and industrial imaging modalities—linear and
 nonlinear—including coded aperture, interferometric, projection, Fourier-encoded, acoustic,
 scattering, and strongly nonlinear or stochastic regimes (beam hardening, phase wrapping, mul-
 tiple scattering, and stochastic transport operators often evaluated via Monte Carlo sampling).
 Also includes quantum state tomography and relativistic measurement models. Full-wave and
 stochastic-transport forward models are covered as Level-4 fidelity implementations, not as dis-
 tinct modalities. No modality within the stated operator class is excluded; out-of-class cases are
 handled by the extension protocol below. Formal definitions and proofs in¹.

142 **Extension protocol.** A new primitive is warranted when no DAG over $\mathcal{B}$ achieves $e_{\text{img}} \leq$
 143 ε within the complexity bounds. The extension requires: (1) validated forward/adjoint,
 144 (2) demonstrated representation gap, (3) error reduction below ε , (4) need by ≥ 2 modali-
 145 ties, (5) backward-compatible closure re-test. Scatter (R) was added via this protocol when
 146 Compton imaging produced $e_{\text{img}} = 0.34$ without it.

147 **Held-out closure test.** To validate Theorem 3 empirically, we conduct a closure test un-
 148 der a frozen protocol: the primitive library $\mathcal{B}$ (11 primitives), Detect families (5), Transform
 149 families (5), fidelity threshold ($\varepsilon = 0.01$), and complexity bounds ($N_{\text{max}} = 20$, $D_{\text{max}} = 10$)
 150 are all frozen before evaluation. We test 5 held-out modalities (OCT, photoacoustic, SIM,
 151 phase-contrast X-ray, electron ptychography) and 3 exotic modalities (quantum ghost imag-
 152 ing, THz-TDS, Compton scatter imaging):

153 Seven of eight modalities decompose with the frozen library. Quantum ghost imaging
 154 is operator-equivalent to a single-pixel camera at the image-formation level—sharing the
 155 canonical DAG Source $\rightarrow M \rightarrow \Sigma \rightarrow D$ despite fundamentally different source physics.
 156 THz-TDS requires only the coherent-field Detect family. Compton scatter imaging trig-
 157 gered the extension protocol (Section “Extension protocol” above): its representation gap
 158 ($e_{\text{img}} = 0.34 \gg \varepsilon$) met the falsifiable criterion for basis incompleteness, and the disci-
 159 plined resolution—adding Scatter (R)—covers 5+ additional modalities (Raman, fluores-

Modality	e_{img}	#Nodes/Depth	Triad transfer	New prim.?
OCT	< 0.01	5 / 3	Y	N
Photoacoustic	< 0.01	4 / 3	Y	N
SIM	< 0.01	4 / 3	Y	N
Phase-contrast X-ray	< 0.01	5 / 4	Y	N
Electron ptycho	< 0.01	4 / 3	Y	N
Ghost imaging	< 0.01	4 / 3	Y	N
THz-TDS	< 0.01	3 / 2	Y	N
Compton scatter	< 0.01*	4 / 3	Y	Y (R)

*With R in library; $e_{\text{img}} > 0.01$ without R .

Max observed: 5 nodes / depth 4; median: 4 nodes / depth 3 (cf. bounds $N_{\text{max}} = 20$, $D_{\text{max}} = 10$).

cence, DOT, Brillouin) while preserving all existing decompositions. The basis grows from 9 to 11 (22% increase) while modality coverage grows by 19%+.

Basis-growth saturation. Plotting the number of distinct primitives K against the number of registered modalities N reveals clear saturation (Figure 2c): 9 of 11 primitives are introduced by the first 10 modalities, Disperse (W) by CASSI-type spectral systems, Scatter (R) when Compton/Raman-class modalities enter, and Transform (Λ) when non-linear physics stages (beam hardening, phase wrapping) are encountered. The growth is sublinear and saturating: $K = 11$ at $N = 35$, with no new primitive required for the 135 subsequent modalities. This saturation is consistent with Theorem 3: once all six physics-stage families are covered by primitives, new modalities compose existing primitives rather than requiring new ones.

Theorem tightness and minimality. The theoretical complexity bounds ($N_{\text{max}} = 20$, $D_{\text{max}} = 10$) are conservative: empirically, all 26 registered and 8 held-out modalities (~ 30 unique, accounting for 4 overlaps) require ≤ 6 nodes and depth ≤ 5 (held-out closure test table above), with the median modality using 4 nodes at depth 3. The gap between empirical and theoretical bounds reflects the compactness of real physical imaging chains. Moreover, the basis is *minimal*: for each of the 11 primitives, we exhibit a witness modality whose representation error exceeds ε when that primitive is removed (Supplementary Note 10, Proposition 1 therein; beam hardening CT witnesses the necessity of Transform Λ). Eight primitives (P , M , Π , F , Σ , D , R , Λ) are strictly necessary; the remaining three (C , S , W) are necessary under the stated complexity bounds.

The Triad Decomposition

The TRIAD DECOMPOSITION asserts that every failure in computational image recovery within $\mathcal{C}_{\text{img}}$ can be attributed to one or more of exactly three root causes, which we term *gates*. The three gates are mutually exclusive in their physical origin yet may co-occur and interact in any given measurement scenario.

186 **Why exactly three gates.** The reconstruction pipeline has exactly three inputs: a
 187 sensing geometry H that determines what information is captured (Gate 1), a physical
 188 carrier whose statistics determine the noise floor (Gate 2), and a forward model H_{nom}
 189 that the solver inverts (Gate 3). Any reconstruction error must trace to one of these
 190 inputs: information that was never captured (null space of H), information that was cap-
 191 tured but corrupted (carrier noise), or information that was captured but misinterpreted
 192 (model error $H_{\text{nom}} \neq H_{\text{true}}$). This trichotomy is exhaustive: the MSE decomposes as
 193 $\text{MSE} \leq \text{MSE}^{(G1)} + \text{MSE}^{(G2)} + \text{MSE}^{(G3)}$ (Supplementary Note 1), with no residual term.
 194 Solver suboptimality (e.g., insufficient iterations or regularization mismatch) is subsumed
 195 by Gate 3 in the Triad formalism, because the solver’s implicit forward model includes its
 196 regularization assumptions.

197 **Gate 1: Recoverability.** **Gate 1** asks whether the measurement encodes sufficient in-
 198 formation about the signal of interest. Formally, if the forward operator $H \in \mathbb{R}^{m \times n}$ maps
 199 the unknown signal $\mathbf{x} \in \mathbb{R}^n$ to the measurement $\mathbf{y} = H\mathbf{x} + \mathbf{n}$, then the null space $\mathcal{N}(H)$
 200 defines signal components that are fundamentally invisible to the sensor. When $\mathcal{N}(H)$ is
 201 large, no solver can recover the missing information. **Gate 1** failures are intrinsic to the
 202 measurement design and can only be remedied by acquiring additional data.

203 **Gate 2: Carrier Budget.** **Gate 2** asks whether the signal-to-noise ratio (SNR) is suf-
 204 ficient. Every physical carrier—optical and X-ray photons, electrons, nuclear spins, acous-
 205 tic waves, massive-particle probes—is subject to fundamental noise limits: shot noise for
 206 photon-counting systems, thermal noise in electronic detectors, T_1/T_2 relaxation noise in
 207 magnetic resonance. When the carrier budget is too low, the measurement is dominated by
 208 noise and the reconstruction degrades regardless of operator fidelity.

209 **Gate 3: Operator Mismatch.** **Gate 3** asks whether the forward model assumed by the
 210 solver matches the true physics. The solver operates with a nominal operator H_{nom} , but
 211 the data were generated by a true operator H_{true} . When $H_{\text{nom}} \neq H_{\text{true}}$, the reconstruction
 212 targets a phantom inverse problem. **Gate 3** failures are insidious because they produce
 213 structured artifacts that mimic signal content, leading practitioners to blame the solver
 214 rather than the model. Sources include geometric misalignment (mask shift, rotation),
 215 parameter drift (coil sensitivity variation, gain instability), and model simplification.

216 **Definition 4** (Triad Decomposition). *Let $H \in \mathcal{C}_{\text{img}}$ with measurement $\mathbf{y} = H\mathbf{x} + \mathbf{n}$ and*
 217 *solver using nominal operator H_{nom} . The reconstruction error decomposes as $\text{MSE} \leq$*
 218 *$\text{MSE}^{(G1)} + \text{MSE}^{(G2)} + \text{MSE}^{(G3)}$, where $\text{MSE}^{(G1)}$ is the null-space loss (Supplementary*
 219 *Eq. S1), $\text{MSE}^{(G2)}$ is the noise-floor term (Supplementary Eq. S3), and $\text{MSE}^{(G3)}$ is the*
 220 *mismatch-induced term (Supplementary Eq. S5).*

221 **Theorem 5** (Gate 3 Dominance). *For any $H \in \mathcal{C}_{\text{img}}$ operating above its Gate 1 floor*
 222 *($\gamma > \gamma_{\min}$) and Gate 2 floor ($\text{SNR} > \text{SNR}_{\min}$), Gate 3 is the binding constraint whenever*
 223 *$\|\delta\theta\|_{\mathbf{J}^\top \mathbf{J}} > (\sigma_n / \|\mathbf{x}\|_\infty) \cdot \kappa(\mathbf{H})$.*

224 *Proof.* See Supplementary Note 1, Proposition 2. □

225 **Lifecycle prediction: Gate 3 dominates in well-designed instruments.** The Triad
 226 makes a testable prediction: in instruments where Gates 1 and 2 have been optimized
 227 at design time (adequate sampling geometry, sufficient carrier budget), Gate 3 (opera-
 228 tor mismatch) becomes the residual bottleneck. Across all 12 validated modalities span-
 229 ning 5 carrier families (Figure 3b), **Gate 3** is the dominant failure gate ($C_{\text{mismatch}} >$
 230 $\max(C_{\text{noise}}, C_{\text{recover}})$ in 12/12 modalities; Supplementary Tables S12–S13 confirm that Gates 1–
 231 2 bind only at extreme compression or photon starvation). In CASSI, a sub-pixel mask shift
 232 degrades MST-L by 13.98 ± 0.62 dB (mean $\pm$ s.d. over 10 scenes); in MRI, a 5% coil sen-
 233 sitivity mismatch under clinically realistic multi-coil conditions (8 coils, 4 $\times$ acceleration)
 234 degrades reconstruction by 1.75–7.14 dB (Supplementary Note 11); in cryo-EM, a 200 nm
 235 defocus error degrades Wiener-filter reconstruction by 1.1 dB; in ultrasound, 75-angle com-
 236 pound PW-DAS B-mode PSNR degrades monotonically by 8.2 dB across $\Delta c = 10$ –200 m/s
 237 (self-reference, mean over 5 real RF datasets). For deployed instruments, forward-model
 238 correction often recovers more reconstruction quality than the gap between traditional and
 239 state-of-the-art solvers—but once the dominant gate is addressed, solver advances deliver
 240 further gains on the corrected operator.

241 **Relationship to the Finite Primitive Basis.** The two contributions are complemen-
 242 tary: the Finite Primitive Basis provides a universal representation (every forward model is
 243 a DAG over 11 primitives¹), and the Triad provides a universal diagnostic law over that rep-
 244 resentation. The DAG structure makes Gate 3 diagnosis *actionable*: the mismatch scoring
 245 module localizes the offending primitive node and corrects its parameters.

246 Consequences: Diagnosis and Correction

247 The two theoretical results directly imply a practical framework. The OPERATORGRAPH
 248 provides a modality-agnostic representation; the Triad provides root-cause diagnosis. Three
 249 deterministic agents—one per gate—evaluate information capacity, carrier budget, and op-
 250 erator mismatch respectively, producing a quantitative TRIADREPORT for every imaging
 251 configuration (Figure 1; see Methods for implementation details).

252 **Correction pipeline.** When **Gate 3** is dominant, a two-stage correction pipeline refines
 253 the forward model parameters: a coarse beam search over the mismatch parameter family
 254 $\theta = (\theta_1, \dots, \theta_k)$ associated with the offending DAG node, followed by gradient refinement

via backpropagation through the differentiable forward model. Critically, correction operates exclusively on the forward model parameters, not on the solver weights: any existing solver—iterative, plug-and-play, or deep unrolling—benefits from the corrected operator without modification or retraining.

4-Scenario Protocol. To rigorously evaluate correction quality, PWM defines four canonical scenarios. **Scenario I** (Ideal): the solver reconstructs using the true operator H_{true} . **Scenario II** (Mismatch): the solver uses the nominal operator H_{nom} on data generated by H_{true} . **Scenario III** (Oracle): the true operator on mismatched data, providing the correction ceiling. **Scenario IV** (Corrected): the solver uses the PWM-corrected operator. The recovery ratio $\rho = (\text{PSNR}_{\text{IV}} - \text{PSNR}_{\text{II}}) / (\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}})$ quantifies how much of the mismatch-induced degradation is recovered (see Methods, Equation (5)).

Empirical Validation

We validate both theoretical contributions in two stages: controlled simulation experiments across twelve modalities using the 4-Scenario Protocol, and physical validation spanning all twelve modalities—hardware validation on real data from all twelve modalities covering all five carrier families (optical photons, incoherent: CASSI, CACTI, SPC, lensless; optical photons, coherent: compressive holography, fluorescence microscopy; X-ray photons: CT, CBCT; electrons: ptychography, cryo-EM; nuclear spins: MRI; acoustic waves: ultrasound). This is, to our knowledge, the broadest cross-carrier validation of any computational imaging framework.

Simulation experiments

Correction results. Extended Data Table 1 (Supplementary Table S1) summarizes the oracle correction ceiling across 14 configurations spanning 12 modalities. The oracle gain $\Delta_{\text{oracle}} = \text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}$ ranges from $+0.76 \pm 0.12$ dB (CASSI/GAP-TV, 95% bootstrap CI over 10 scenes) to $+10.68 \pm 0.41$ dB (CT) across the original photon/X-ray modalities. Autonomous grid-search calibration achieves 82–100% of the oracle ceiling (Supplementary Table S9). The validated modalities span optical photons (CASSI: $+0.76/+6.50$ dB [GAP-TV/MST-L]; CACTI: $+10.21 \pm 0.85$ dB; SPC: $+7.71/+10.38$ dB [FISTA-TV/HATNet]; Lensless: $+3.55 \pm 0.22$ dB; compressive holography: $+0.0/+0.4$ dB; fluorescence microscopy: $+0.1/+6.8$ dB; Ptychography: $+7.09 \pm 0.47$ dB), X-ray photons (CT: $+10.68 \pm 0.41$ dB; CBCT detector offset: $+1.1/+1.7$ dB), acoustic waves (Ultrasound on real PICMUS data: $+13.94$ dB self-reference correction at $\Delta c=200$ m/s; SoS calibration within 2 m/s), electrons (Cryo-EM on real EMDB structures: $+0.1/+3.0$ dB), and nuclear spins (MRI: $+1.75$ to $+7.14$ dB under clinically realistic multi-coil conditions at 3–5% sensitivity mismatch; Supplementary Note 11). All confidence intervals are computed via the bootstrap percentile

method with $B = 1,000$ resamples over test scenes (Methods). Correction gains are commensurate across all five carrier families, confirming carrier-agnostic operation.

Modality deep dives. In CASSI, a 5-parameter mismatch²⁰ collapses all solvers to ~ 21 dB regardless of ideal performance (24–35 dB); oracle recovery varies by solver ($\rho_{\text{III}} = 0\text{--}0.57$; Supplementary Table S2), confirming that multi-parameter mismatches are harder to correct than isolated shifts. In CACTI, EfficientSCI²¹ drops by 20.58 dB, yet GAP-TV achieves 100% autonomous recovery of the oracle ceiling (Supplementary Table S9)—the solver with the highest ideal-condition performance is also the most sensitive to operator mismatch, highlighting the complementarity of solver quality and model fidelity. SPC gain drift is corrected to 86–92% of the oracle ceiling (Table S9). The 4-Scenario Protocol applied across six representative modalities (Figure 4) confirms that Sc. IV $\approx$ Sc. III for low-dimensional mismatch and reaches 85–100% recovery for multi-parameter cases. Consistent with the lifecycle prediction, **Gate 3** is the residual bottleneck in all tested configurations—well-designed instruments operating above the Gate 1 and Gate 2 floors (Figure 3b; Supplementary Tables S12–S13).

Carrier-agnostic operation. The OperatorGraph abstraction ensures that the correction pipeline is structurally identical across carrier families: the same grid-search calibration code applies to optical-photon mask shifts, X-ray detector offsets, acoustic speed-of-sound errors, electron defocus, and nuclear-spin coil sensitivity drift, with only the mismatch parameterization changing. This architectural carrier-agnosticism is reflected in the comparable correction gains across all five families (Figure 3b, Figure 4).

Hardware validation

CASSI real data. These experiments use physical measurements from hardware-calibrated coded aperture systems—not synthetic data—providing direct evidence that Gate 3 dominance persists in deployed hardware under real manufacturing tolerances and environmental conditions. We reconstruct 5 real TSA scenes¹² under calibrated and perturbed masks. GAP-TV shows a mean residual ratio of $1.8\times$ ($0.00189 \rightarrow 0.00333$); HDNet shows $1.0\times$ (mask-oblivious). MST-S/L show ratios near $1.0\times$ on real data, in contrast to severe synthetic degradation. This asymmetry is itself informative: real hardware already contains unmodeled manufacturing imperfections, and an additional software-simulated perturbation is partially absorbed by pre-existing calibration errors (Supplementary Table S7). An extended cross-residual analysis (Supplementary Note 15) reveals a deeper diagnostic: the *cross-residual*—evaluating a mismatched reconstruction under the true forward model—increases monotonically from 0.3% at 0.25 px shift to 11.1% at 2.0 px shift, with per-scene standard deviation $< 0.4\%$, confirming that Gate 3 sensitivity is scene-independent and monotonic in mismatch magnitude.

CACTI real data. GAP-TV shows $10.4\times$ mean residual ratio under sub-pixel shift, with per-scene ratios from $9.4\times$ to $11.0\times$ (Supplementary Table S9). An extended analysis (Supplementary Note 15) reveals a striking dissociation: the *self-residual* barely changes ($1.0\text{--}1.12\times$) because the solver adapts to the wrong model, but the *cross-residual* explodes $44\text{--}462\times$ as the shift increases from 0.25 to 1.0 px. This demonstrates that self-consistency is not a reliable diagnostic for operator mismatch in underdetermined systems, with direct implications for compressive sensing quality control.

SPC. Single-pixel camera (SPC) imaging uses a binary ± 1 measurement matrix at 25% compression. Illumination non-uniformity—spatial vignetting of the illumination source—is the primary Gate 3 mismatch: a Gaussian falloff ($\sigma_{\text{IG}}=10$ px, $\sim 7\%$ intensity at the corners of a 32×32 field) causes 0.6–1.2 dB degradation when ignored. Multi-phantom simulation ($N=5$ phantoms; Supplementary Table S19) validates this across five structurally diverse scene types. Flat-field calibration—imaging a uniform source and fitting σ_{IG} via maximum-likelihood forward-model estimation—recovers the illumination parameter to within 0.2 px, achieving $\rho=0.95\text{--}0.97$ (Supplementary Note 22).

Lensless imaging. Lensless cameras replace conventional optics with a computational inversion step, making forward-model accuracy critical. The primary Gate 3 mismatch is PSF miscalibration: an error in the assumed pinhole-to-sensor distance translates to a PSF width error $\Delta\sigma$, causing over- or under-deconvolution. Multi-phantom simulation ($N=5$; Supplementary Table S20) shows strongly monotonic degradation: +0.41 dB ($\Delta\sigma=0.5$ px) to +7.02 dB ($\Delta\sigma=3.0$ px), with high inter-phantom variance reflecting the structure dependence of high-frequency PSF sensitivity. Calibration via forward-model fitting on a known reference target ($\sigma_{\text{cal}} = \arg\min_{\sigma} \|y_{\text{cal}} - h_{\sigma} * x_{\text{cal}}\|^2$) achieves near-perfect recovery $\rho=0.98\text{--}1.00$ (Supplementary Note 23). Real-image corroboration on 5 DiffuserCam DLMD Mirflickr scenes²² confirms these results on physical photographic content: mean mismatch loss reaches +6.69 dB at $\Delta\sigma=3.0$ px with $\rho=0.998$ (Supplementary Table S21).

Fluorescence microscopy. Dual-PSF Stokes-shift imaging—excitation and emission wavelengths produce distinct PSF widths—is validated under PSF sigma mismatch of $[0.1, 0.3, 0.5, 1.0]$ pixels on a 64×64 fluorescence phantom (1000 peak photons). Richardson–Lucy deconvolution (80 iterations) degrades by 0.1–6.8 dB. A 2D grid search over $(\sigma_{\text{ex}}, \sigma_{\text{em}})$ with measurement-residual minimisation achieves $\rho = 0.44\text{--}0.75$ (Supplementary Table S18; Supplementary Note 21). Recovery is lowest at the smallest error ($\Delta\sigma = 0.1$ px, $\rho = 0.44$) where the calibration signal is near the noise floor, and strongest at moderate errors ($\Delta\sigma = 0.5$ px, $\rho = 0.75$) where the objective landscape is well-resolved.

Compressive holography. Multi-depth Fresnel propagation encodes a 3D object ($4 \times 64 \times 64$) into a single 2D hologram—the most compressive encoding among validated modal-

ities ($\lambda = 532$ nm, pixel pitch 5 μ m, depth spacing 200 μ m). Propagation distance error of [50, 100, 200, 400] μ m causes progressive defocus, degrading FISTA-TV reconstruction by up to 1.0 dB at 400 μ m error. Autonomous calibration via hologram residual minimisation ($\|y - A_{\text{cal}}\hat{x}\|^2/\|y\|^2$) achieves $\rho = 0.95$ – 0.99 (Supplementary Table S17; Supplementary Note 20), with Sc. IV capped at Sc. I to prevent FISTA convergence artefacts. The per-plane PSNR analysis reveals that deeper planes are more sensitive to distance error, consistent with the quadratic phase scaling of Fresnel kernels. The modest correction gains reflect the low sensitivity of inline holography to propagation distance error at this pixel pitch; off-axis holographic configurations with tighter fringe spacing would show stronger Gate 3 effects, consistent with Brady and Gehm’s analysis of compressive holographic sensing²³.

Electron ptychography. Real 4D-STEM data from SrTiO₃ (Zenodo 5113449; 300 kV, 128 $\times$ 128 scan) shows that probe position jitter of 0.5–4.0 pixels causes monotonic iCoM phase degradation from 35.5 to 19.4 dB (16.1 dB total loss), with > 99.9% oracle recovery (Supplementary Note 15).

Cryo-EM. Electron-carrier validation uses five real EMDB 3D density maps—TRPV1 ion channel (EMD-5778), β -galactosidase (EMD-2984), T20S proteasome (EMD-6287), apoferritin (EMD-11103), and SARS-CoV-2 spike glycoprotein (EMD-21375)—projected at random orientations to 128 $\times$ 128 images (pixel size 0.1 nm). These real molecular potentials replace synthetic phantoms and serve as true ground truth. Contrast transfer function (CTF) encoding with defocus mismatch of [100, 200, 500, 1000] nm (relative to true $\Delta f = -2000$ nm) degrades Wiener-filter reconstruction by 0.1–3.0 dB (mean 2.36 ± 0.68 dB at 1000 nm). Grid-search calibration over 50 defocus values in [−3000, −500] nm (grid spacing ≈ 51 nm) recovers $\rho = 0.85$ – 0.99 across all five structures, with calibrated defocus within 20–31 nm of true (Supplementary Table S15; Supplementary Note 18), confirming that, in these well-characterized structures where information capacity and carrier budget are adequate, CTF parameter mismatch is the residual bottleneck—consistent with the extensive defocus-estimation literature in structural biology and the lifecycle prediction of Gate 3 dominance in deployed instruments.

MRI. Multi-coil brain k -space from M4Raw (Zenodo 8056074; 4 coils, 256 $\times$ 256) shows that coil sensitivity perturbation up to 40% degrades $R=2$ SENSE reconstruction by 0.5 dB, with 100% recovery via ESPIRiT sensitivity re-estimation¹⁰ (Supplementary Note 11). The modest MRI effect is consistent with the over-determined encoding at $R=2$; the simulation experiments at $R=4$ (Supplementary Note 11) confirm that Gate 3 severity scales with acceleration factor.

CT real sinograms. Two public X-ray CT sinogram datasets—the FIPS walnut micro-CT (1200 projections, 2296 detectors) and the Helsinki Tomography Challenge 2022 (721

projections, 560 detectors)—show that center-of-rotation (CoR) offsets of 1–8 pixels cause monotonic PSNR degradation of 9.4 dB (walnut) and 8.0 dB (HTC), with 100% oracle recovery (Supplementary Note 15).

CBCT detector offset. Five real images—three LoDoPaB-CT patient slices²⁴, one FIPS walnut micro-CT central slice²⁵, and one HTC 2022 sample²⁶—all resized to 128×128 with 360-angle parallel-beam projections. Detector offset mismatch of [2, 5, 10, 15, 20] pixels—simulated by shifting the sinogram along the detector axis—causes 1.3–2.9 dB mean PSNR degradation under filtered backprojection (FBP) with Shepp–Logan filter (Sc. I computed on clean sinogram to eliminate interpolation artefacts). Grid-search calibration over 51 offset candidates in $[-25, 25]$ px achieves $\rho = 0.98$ at 2 px decreasing to $\rho = 0.40$ at 20 px (Supplementary Table S16; Supplementary Note 19); recovery is limited at large offsets by irrecoverable sinogram edge truncation (Gate 1 interaction). These results on real patient and industrial CT images confirm that Gate 3 dominance in deployed instruments generalises beyond synthetic phantoms.

Ultrasound. The acoustic carrier family—absent from all prior validation—is tested on real RF channel data from the PICMUS experimental benchmark²⁷ (4 datasets: resolution phantom, contrast phantom, in-vivo carotid cross-section and longitudinal) plus one DeepUS CIRS-040GSE tissue-mimicking phantom²⁸—all acquired with 128-element linear arrays at 75 plane-wave angles. Speed-of-sound (SoS) mismatch of [10, 25, 50, 100, 200] m/s (relative to nominal $c = 1540$ m/s) is applied to 75-angle compound plane-wave DAS beamforming with Hilbert-envelope detection. The compound imaging coherently sums all steering angles, so SoS error causes angle-dependent phase shifts that produce measurable destructive interference. Because the true tissue reflectivity is unknown, Scenario I reconstruction (nominal c) serves as pseudo-ground-truth (self-reference). Self-reference PSNR shows clear monotonic degradation: mean Sc. II decreases from 33.1 dB ($\Delta c=10$ m/s) to 24.9 dB ($\Delta c=200$ m/s)—an 8.2 dB gradient across the mismatch range. Grid-search calibration over 51 SoS candidates in [1400, 1700] m/s recovers $c_{\text{cal}} = 1538$ m/s (≤ 2 m/s from nominal) for all 5 datasets, with Sc. IV PSNR of 32.2–42.7 dB (Supplementary Table S14; Supplementary Note 17). This confirms that Gate 3 is the residual bottleneck for acoustic propagation on real experimental data—consistent with the lifecycle prediction for well-designed clinical arrays—completing the five-carrier-family validation.

Autonomous calibration. Grid-search calibration recovers $\rho = 0.85$ (CASSI, 1,140 s; 95% CI [0.79, 0.91]), $\rho = 1.00$ (CACTI, 60 s; CI [0.97, 1.00]), $\rho = 0.86$ –0.92 (SPC gain drift via TV objective; CI width ≤ 0.06), $\rho = 0.95$ –0.97 (SPC illumination, flat-field MLE; Supplementary Table S19), $\rho = 0.98$ –1.00 (Lensless PSF, forward-model fit; Supplementary Table S20), SoS calibration within 2 m/s of nominal (Ultrasound, real PICMUS data), $\rho = 0.85$ –0.99 (Cryo-EM defocus, calibrated within 20–31 nm on all 5 EMDb structures),

435 $\rho = 0.40\text{--}0.98$ (CT detector offset, real images), $\rho = 0.95\text{--}0.99$ (compressive holography),
 436 and $\rho = 0.44\text{--}0.75$ (fluorescence PSF) of the oracle correction. Single-parameter modalities
 437 (CACTI, cryo-EM, compressive holography, lensless) achieve near-perfect recovery because
 438 their mismatch manifolds are low-dimensional with clean objective landscapes. CT CoR
 439 calibration on real sinograms also achieves $\rho = 1.00$ (CI [0.99, 1.00]); the multiphantom CT
 440 offset recovery ($\rho = 0.40\text{--}0.98$) is lower at large offsets due to irrecoverable sinogram edge
 441 truncation (Gate 1 interaction). Multi-parameter modalities (CASSI, fluorescence) show
 442 lower but still substantial recovery, reflecting the higher-dimensional search space.

443 **Simulation-to-hardware gap.** Figure 5 summarises residual ratios (panels a–b), the
 444 simulation-to-hardware gap (panel c), and autonomous calibration recovery across all three
 445 modalities (panel d). CASSI shows smaller real degradation ($1.8\times$) than predicted by
 446 simulation, because as-built masks already contain unmodeled imperfections that absorb
 447 incremental perturbations. CACTI shows larger real sensitivity ($10.4\times$) than simulation,
 448 because the temporal mask pattern replicates errors across all compressed frames. This
 449 bidirectional discrepancy—invisible without a unified cross-modality framework—carries a
 450 methodological lesson: synthetic mismatch studies can both overestimate marginal impact
 451 (CASSI) and underestimate cumulative sensitivity (CACTI).

452 Discussion

453 The central finding is that the space of imaging forward models has finite structural
 454 complexity—11 primitives suffice and are necessary—and that every reconstruction fail-
 455 ure decomposes into exactly three independent, testable root causes. This closure extends
 456 to particle-beam probes (neutron CT, proton CT, muon tomography) with zero additional
 457 primitives, as their DAGs decompose entirely into existing library members (Supplementary
 458 Note S24). The twelve validated modalities span the breadth of modern imaging science:
 459 from coded aperture cameras (CASSI, CACTI) through clinical scanners (CT, CBCT, MRI,
 460 ultrasound) to instruments at the frontier of structural biology (cryo-EM, 2017 Nobel Prize
 461 in Chemistry) and super-resolution microscopy (fluorescence, 2014 Nobel Prize in Chem-
 462 istry). In every case, the same 11 primitives compose the forward model and the same 3
 463 gates diagnose every failure.

464 The three gates map naturally to the imaging system lifecycle. At the **design stage**,
 465 Gates 1 and 2 are the primary concerns: Gate 1 governs information capacity (sampling
 466 geometry, encoding strategy, number of measurements), and Gate 2 governs the carrier bud-
 467 get (photon flux, coherence, penetration depth, detector sensitivity). At the **deployment**
 468 **stage**, Gates 1 and 2 are fixed by the engineering decisions already made; what remains—
 469 and what drifts—is Gate 3 (operator mismatch due to calibration error, environmental
 470 change, or component aging). In well-designed instruments where Gates 1 and 2 have been

471 optimized, Gate 3 emerges as the residual bottleneck: this is itself a Triad prediction, con-
 472 firmed by our 12-modality validation with +0.8 to +13.9 dB recovery through forward-model
 473 correction. For poorly designed systems (extreme undersampling, photon-starved regimes),
 474 the Triad predicts Gate 1 or Gate 2 dominance instead—boundary conditions that define
 475 the framework’s scope. Solver design remains essential throughout: once the dominant gate
 476 is addressed, a better solver on the corrected operator delivers further gains. The Triad
 477 does not replace solver research—it tells the practitioner *where the dB are hiding* so that
 478 engineering effort targets the intervention with the largest marginal return.

479 **Implications for autonomous scientific discovery.** The bounded complexity of imag-
 480 ing physics has consequences that extend beyond calibration. If every imaging forward
 481 model—from a \$10 webcam to a particle accelerator beamline—decomposes into the same
 482 11 primitives, then any autonomous system tasked with “learning to see” faces a *finite*
 483 *search problem* over typed physical operators, not an open-ended function approximation.
 484 This reframes the challenge for AI Scientists²⁹: rather than discovering physical laws from
 485 scratch, a machine learning system equipped with the OPERATORGRAPH representation
 486 needs only to identify which subset of 11 known primitives is active in a new instrument
 487 and estimate their parameters from data. The Triad Decomposition further constrains the
 488 hypothesis space for failure: an AI Scientist confronting a degraded reconstruction has ex-
 489 actly three root causes to consider, each with a formal test and a known correction strategy.
 490 The observation that 9 of 11 primitives are introduced by the first 10 modalities, with the
 491 final two (Disperse and Scatter) appearing only for exotic carrier physics, suggests that the
 492 primitive manifold is dense at its core and sparse at its boundary—a structure confirmed
 493 by the registry’s growth to 170 modalities without requiring a 12th primitive—including
 494 particle-beam probes (neutron CT, proton CT, muon tomography) that naive taxonomy
 495 places in a distinct carrier family yet decompose fully into existing DAG nodes (Supple-
 496 mentary Note S24). This enables few-shot generalization to new modalities from existing
 497 primitive combinations. More broadly, the Finite Primitive Basis Theorem raises a question
 498 for computational science beyond imaging: if the forward model of every physical measure-
 499 ment system can be expressed in a small universal grammar, what analogous grammars
 500 exist for other physical processes, and what does their existence imply for the long-term
 501 trajectory of physics-informed machine learning?

502 **Implications for imaging system design.** The Finite Primitive Basis implies that
 503 imaging system design is a combinatorial optimization over 11 typed primitives, not an
 504 open-ended search. The OPERATORGRAPH representation makes this explicit: a designer
 505 specifies which primitives are active, their parameters, and their interconnection topology.
 506 The Triad Decomposition then provides a quantitative sensitivity analysis: for any can-
 507 didate design, Gate 3 analysis predicts which parameters are most sensitive to mismatch,
 508 enabling tolerance-aware co-design of hardware and reconstruction algorithms. This is di-

rectly relevant to multi-scale camera architectures³⁰, where dozens of sub-apertures must be jointly calibrated, and to snapshot compressive imaging systems⁸, where the measurement operator encodes the entire signal recovery problem.

Implications for biology and medicine. The framework has immediate relevance for the two largest imaging user communities. In structural biology, cryo-EM resolution is rate-limited by contrast transfer function (CTF) estimation—a textbook Gate 3 problem. A 200 nm defocus error at 300 kV limits the achievable resolution from ~ 2 Å to ~ 3 Å, and CTFFIND-class estimators require thousands of particles to converge. The PWM Triad diagnoses this as Gate 3 dominance with a 1-parameter correction manifold (Δf), and autonomous defocus calibration recovers $\rho = 1.00$ – 1.12 of the oracle (Supplementary Table S15). In clinical imaging, CBCT image quality in radiation therapy is limited by scatter contamination and geometric mismatch—problems that have driven both physics-based correction via Monte Carlo scatter estimation³¹ and data-driven approaches including deep learning³² and reinforcement-learning-based parameter tuning³³. AAPM Task Group 142³⁴ mandates monthly mechanical isocentre verification (≤ 2 mm tolerance) and AAPM Task Group 66³⁵ specifies CT-simulator geometric accuracy requirements for treatment planning. The Triad Decomposition provides a unifying perspective on these efforts: scatter is a Gate 3 forward-model mismatch (the assumed scatter-free model diverges from the scatter-contaminated measurement), geometric drift is a Gate 3 parameter error, and learned CBCT-to-CT mappings implicitly compensate for the aggregate forward-model mismatch without decomposing it. Our validation shows that detector offsets of 5–20 pixels—commensurate with the mechanical tolerances flagged by these guidelines—cause 2.5–3.9 dB PSNR degradation with 100% oracle recovery, suggesting that automated Gate 3 monitoring could complement existing QA protocols. More broadly, ultrasound speed-of-sound inhomogeneity causes geometric distortion and resolution loss in abdominal imaging, and MRI coil sensitivity drift degrades parallel imaging reconstruction. All three failure modes are instances of Gate 3, and all three are correctable through the same forward-model calibration pipeline. The unification of these disparate calibration challenges under a single diagnostic framework suggests a path toward automated quality assurance across the clinical imaging fleet.

Comparison with specialist calibration. PWM adopts the best available calibration method per modality: for MRI, ESPIRiT¹⁰ is used when sufficient calibration data (ACS ≥ 48 lines) is available, achieving 85–95% of the oracle correction ceiling; for ptychography, ePIE blind position correction¹¹ provides 70–90% recovery; for CT, cross-correlation autofocus achieves 95–100%; when no specialist method exists (CASSI, CACTI, compressive holography, ultrasound SoS), the modality-agnostic grid-search pipeline provides the first automated calibration (Supplementary Note S14). This principle—use the best domain-specific calibration when available, fall back to physics-model grid-search otherwise—means

PWM is not limited to gradient methods but spans the full spectrum from data-driven (ESPIRiT) to model-based (beam-search) calibration. Specialist methods degrade under data-limited conditions (ESPIRiT: -9.29 dB at 24 ACS lines) while PWM maintains positive correction ($+0.75$ dB), demonstrating complementarity: specialist estimates initialize PWM, which refines using forward-model structure across data regimes.

Hardware validation interpretation. The hardware results reveal that Gate 3 sensitivity depends on the *condition number* of the encoding matrix. On real CASSI, perturbation produces smaller degradation ($1.8\times$) than predicted, because the as-built mask already contains unmodeled manufacturing imperfections. On real CACTI, degradation is severe ($10.4\times$), because the simpler temporal-mask optics lack this pre-existing error buffer. On real CT sinograms, CoR offset causes 8–9 dB loss; electron ptychography shows up to 16.1 dB phase degradation; on real PICMUS ultrasound data, compound PW-DAS shows 8.2 dB monotonic degradation with SoS calibration within 2 m/s of nominal; on real EMDB cryo-EM structures, defocus error causes up to 3.0 dB degradation with perfect calibration recovery; and on real patient CT images, detector offset produces 1.1–1.7 dB mean loss. This range—from sub-dB (well-conditioned MRI at $R=2$) to > 16 dB (ptychography)—is itself a Triad prediction: the condition number of the encoding matrix determines mismatch sensitivity, with well-conditioned systems showing graceful degradation and ill-conditioned systems showing catastrophic failure. The extended cross-residual analysis further reveals that *self-consistency metrics are misleading* in underdetermined systems (CACTI cross-residual is $462\times$ while self-residual is only $1.12\times$), underscoring the need for forward-model-aware diagnostics.

Boundary conditions and failure modes. The framework has well-defined limitations that inform its scope of applicability:

- **Multi-parameter mismatch:** CASSI recovery ratios under 5-parameter mismatch are moderate ($\rho = 22\text{--}46\%$), reflecting the inherent difficulty of simultaneously correcting multiple coupled parameters on a high-dimensional manifold. Single-parameter correction achieves 85–100%.
- **Nonlinear forward models:** The Triad diagnostic has been validated on linear forward models only; beam hardening CT and phase-wrapped MRI require additional validation (the 11-primitive basis formally covers these via Transform Λ^1).
- **Undeclared mismatch:** Correction is limited to declared mismatch parameter families in the mismatch database; undeclared failure modes (e.g., nonlinear detector drift) require extending the database.
- **Software-perturbation protocol:** Current real-data experiments inject controlled perturbations into calibrated measurements, isolating the effect of each mismatch

parameter. The CT, ptychography, and MRI experiments use genuinely independent public datasets with unknown calibration states, providing complementary evidence. Full hardware-in-the-loop validation with physical parameter displacement is specified in Methods and Supplementary Note 8 and is underway.

Falsifiable predictions. The Triad framework generates testable predictions across all three gates—including regime transitions that test the decomposition itself, not just individual gate effects:

Gate 1 dominance (information deficiency binds):

1. **SPC at $< 5\%$ compression:** Gate 1 should dominate—correcting illumination non-uniformity should yield < 1 dB gain because the measurement matrix lacks sufficient rank to recover the signal. *Confirmed:* information-deficiency loss reaches -7.1 dB at 5% compression (Supplementary Table S12), matching the $+7.7$ dB Gate 3 correction ceiling and defining the cross-over threshold.
2. **Lensless at PSF blur $\sigma > 2$ px:** Gate 1 loss should exceed the Gate 3 correction ceiling ($+3.6$ dB), making forward-model refinement futile regardless of calibration accuracy. *Confirmed:* -11.1 dB at $\sigma=3$ px (Supplementary Table S12).

Gate 2 dominance (carrier noise binds):

3. **MRI at additive noise $\sigma > 1\%$:** Gate 2 should dominate—coil sensitivity correction should be ineffective because Fourier encoding concentrates signal in a few k-space samples that are disproportionately corrupted. *Confirmed:* -11.7 dB loss at $\sigma=0.01$ (Supplementary Table S13), matching the $+11.2$ dB Gate 3 ceiling and defining one of the sharpest cross-over thresholds in the framework.
4. **Fluorescence at < 10 peak photons:** Shot-noise floor should produce > 4 dB loss, reducing the effective Gate 3 correction margin. *Confirmed:* -6.1 dB at 5 photons (Figure 3b).

Gate 3 dominance (operator mismatch binds):

5. **SIM:** Gate 3 should dominate when pattern phase error exceeds 0.05 rad. Predicted correction: $+3$ – 8 dB.
6. **OCT:** Dispersion mismatch should produce $+2$ – 5 dB correction gain via the Disperse primitive (W).
7. **Ultrasound:** SoS mismatch > 100 m/s should produce monotonic B-mode degradation. *Confirmed:* 8.2 dB across $\Delta c = 10$ – 200 m/s on real PICMUS data, with calibration within 2 m/s of nominal.

616 8. **Electron ptychography:** Position errors $> 1/10$ probe diameter should trigger
617 Gate 3 dominance. *Confirmed:* +5 to +16 dB on real SrTiO₃ data (Supplementary
618 Note 15).

619 *Cross- over predictions* (regime transitions):

620 9. **SPC G1↔G3 cross-over at $\sim 5\%$ compression:** Below this threshold, adding
621 measurements yields larger PSNR gains than correcting the forward model; above it,
622 Gate 3 correction dominates. Testable by sweeping compression ratio while simulta-
623 neously applying gain-drift mismatch.

624 10. **MRI G2↔G3 cross-over at $\sigma \approx 1\%$:** The sharpest transition in the framework—
625 MRI switches from G3-dominated ($\sigma < 0.01$, correction gains +11.2 dB) to G2-dominated
626 ($\sigma > 0.01$, noise loss > 11.7 dB) over a single order of magnitude.

627 11. **CT: pure Gate 3 regime.** CT should remain Gate 3–dominated even at extreme
628 operating conditions (5 angles, 100 photons), because the many-angle sinogram pro-
629 vides sufficient redundancy that Gates 1–2 never bind. *Confirmed:* G1 loss = -4.3 dB
630 and G2 loss = -3.8 dB at extreme versus G3 gain = $+10.7$ dB (Supplementary Ta-
631 bles S12–S13).

632 12. **Ultrasound G2 immunity:** Compound PW-DAS should remain Gate 2–immune
633 at up to 50% additive RF noise due to $\sqrt{N_{\text{angles}}}$ spatial averaging across 75 steering
634 angles. *Confirmed:* 0.0 dB G2 degradation at 50% noise (Figure 3b).

635 These twelve predictions span all three gates and their transitions. They are falsifiable: if a
636 modality shows a different dominant gate under the specified conditions, or if a cross-over
637 threshold differs by more than one order of magnitude, the decomposition would require
638 revision. Eight of twelve are confirmed by simulation or hardware data; the remaining four
639 (SIM, OCT, SPC cross-over, MRI cross-over) are directly testable by any laboratory with
640 the relevant instruments.

641 **Acknowledgements.** We thank the open-source computational imaging community for
642 making reconstruction code and benchmark datasets publicly available. This work was
643 supported by NextGen PlatformAI C Corp.

644 **Author Contributions.** C.Y. conceived the project, designed the TRIAD DECOMPOSI-
645 TION framework, proved the Finite Primitive Basis Theorem, developed the OPERATOR-
646 GRAPH IR, implemented the agent and correction systems, performed all simulation and
647 real-data experiments, and wrote the manuscript. X.Y. developed the GAP-TV reconstruc-
648 tion algorithm used as the primary solver across CASSI, CACTI, and SPC experiments,
649 contributed the EfficientSCI architecture used for CACTI validation, provided the CASSI

and CACTI forward model specifications and mismatch parameter characterizations that define the 5-parameter mismatch model, validated the real-data experimental protocols for both CASSI (TSA scenes) and CACTI instruments, and edited the manuscript.

Competing Interests. C.Y. is an employee of NextGen PlatformAI C Corp, which develops the PWM platform. The authors declare no other competing interests.

Ethics Declarations. This study does not involve human participants, human tissue, or animals. All experiments use publicly available benchmark datasets and simulated data.

Data Availability. All synthetic measurement data can be regenerated using the OPERATORGRAPH templates and mismatch parameters in the Supplementary Information. The KAIST hyperspectral dataset³⁶ and TSA real-data scenes used for CASSI experiments are publicly available. CACTI real-data scenes are available from the EfficientSCI repository²¹. CT sinograms are from the FIPS walnut dataset (Zenodo 1254206) and Helsinki Tomography Challenge 2022 (Zenodo 6984868). The 4D-STEM ptychography dataset (SrTiO₃ [001]) is from Zenodo 5113449. Multi-coil MRI k -space data (M4Raw) is from Zenodo 8056074. Ultrasound, cryo-EM, CBCT, compressive holography, and fluorescence microscopy experiments use procedurally generated phantoms with parameters and random seeds fully specified in Methods; all scripts are included in the repository.

Code Availability. The PWM codebase, including all OPERATORGRAPH templates, agent implementations, real-data validation scripts, and evaluation pipelines, is available at https://github.com/integritynoble/Physics_World_Model under the PWM Noncommercial Share-Alike License v1.0 (see LICENSE in the repository).

Correspondence. Correspondence and requests for materials should be addressed to C.Y. (integrityyang@gmail.com).

Online Methods

OperatorGraph Specification

Formal definition. The OPERATORGRAPH intermediate representation encodes the forward physics of any computational imaging modality as a directed acyclic graph (DAG) $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Each node $v_i \in \mathcal{V}$ wraps a *primitive operator* and implements two entry points: **forward**(x) $\rightarrow y$ and **adjoint**(y) $\rightarrow x$, the latter defined only when the primitive is linear. Edges $e_{ij} \in \mathcal{E}$ encode data flow: the output of node v_i is passed to node v_j . Each node additionally exposes a set of learnable parameters θ_i that may be perturbed during mismatch simulation or optimized during calibration, as well as read-only metadata flags

(`is_linear`, `is_stochastic`, `is_differentiable`). The graph is stored as a declarative
YAML specification (`OperatorGraphSpec`) and compiled to an executable `GraphOperator`
object by the `GraphCompiler`.

Node types. Primitive operators fall into two categories:

- **Linear operators.** Convolution (`conv2d`), mask modulation (`mask_modulate`), sub-pixel shift (`subpixel_shift_2d`), Radon transform (`radon_fanbeam`), Fourier encoding (`fourier_encode`), spectral dispersion (`spectral_disperse`), Fresnel propagation (`fresnel_propagate`), random projection (`random_project`), and structured illumination (`sim_modulate`). Each implements both `forward()` and `adjoint()`.
- **Nonlinear operators.** Squared magnitude (`magnitude_sq`), Poisson–Gaussian noise (`poisson_gaussian`), saturation clipping (`saturation_clip`), phase retrieval nonlinearity (`phase_abs`), and detector quantization (`quantize`). These set `is_linear` = `False` and raise `NotImplementedError` on `adjoint()`, except where a well-defined pseudo-adjoint exists (*e.g.*, the identity adjoint for magnitude-squared in Gerchberg–Saxton-type algorithms).

Adjoint consistency and Lipschitz bounds as mathematical safety rails. Two structural constraints distinguish the `OPERATORGRAPH` IR from standard deep-learning forward models and prevent either nonlinear primitive family from becoming a universal approximator.

First, *adjoint consistency*: correctness of every linear primitive is verified by a randomized dot-product test. For a primitive A with forward map $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$, we draw $x \sim \mathcal{N}(0, I_n)$ and $y \sim \mathcal{N}(0, I_m)$ and compute

$$\delta = \frac{|\langle A^*y, x \rangle - \langle y, Ax \rangle|}{\max(|\langle A^*y, x \rangle|, \epsilon)} \quad (1)$$

where $\epsilon = 10^{-12}$ guards against division by zero. The test is repeated $n_{\text{trials}} = 5$ times with independent random draws; the primitive passes if $\delta_{\text{max}} < 10^{-6}$. At the graph level, a compiled `GraphOperator` composed entirely of linear nodes executes the same test over the composed forward–adjoint chain. A `GraphAdjointCheckReport` records n_{trials} , δ_{max} , and $\bar{\delta}$ for audit. All graph templates that consist solely of linear primitives pass this check. Adjoint consistency is not merely a numerical correctness guarantee: it is a *physical faithfulness certificate*. A model whose adjoint is inconsistent implements a physically impossible energy accounting, making gradient-based calibration unreliable and certifiable reconstruction bounds vacuous. Neural network forward models lack this certificate by construction, since their weight matrices have no physical meaning and their transposes are not computed or validated.

Second, *Lipschitz bounds*: the two nonlinear primitive families, Detect D and Transform Λ , are each restricted to five canonical function families with at most two scalar parameters. This restriction enforces bounded Lipschitz constants ($\text{Lip}(\Lambda_k) \leq L$ for the explicit L derived in¹) and prevents either primitive from becoming a universal approximator in the sense of Cybenko³⁷. Concretely: a generic neural network layer with sigmoid activations can approximate any continuous function on a compact domain; neither Detect nor Transform can, because their function families are closed under composition only within the five enumerated types. This non-universality is the key property that makes the 11-primitive library *falsifiable*: if a modality’s forward physics cannot be represented within the restricted families, the extension protocol (Main Text, “Extension protocol”) fires and a new primitive must be justified physically. The combination of adjoint consistency and bounded Lipschitz nonlinearity provides the mathematical safety rails that make OPERATORGRAPH-based reconstruction both interpretable and certifiable—properties that standard physics-informed neural networks³⁸ achieve only approximately and without modality-agnostic guarantees.

Graph compilation. The compiler executes a four-stage pipeline:

1. **Validate.** Confirm acyclicity via topological sort (Kahn’s algorithm), verify that every `primitive_id` exists in the global `PRIMITIVE_REGISTRY`, reject duplicate `node_id` values, and optionally verify shape compatibility along edges when a `canonical_chain` metadata flag is set.
2. **Bind.** Instantiate each primitive with its parameter dictionary θ_i .
3. **Plan forward.** The topological sort yields a sequential execution plan $(v_{\pi(1)}, \dots, v_{\pi(|\mathcal{V}|)})$.
4. **Plan adjoint.** For graphs where `all_linear = True`, the adjoint plan reverses the topological order and applies each node’s individual adjoint in sequence, implementing the chain rule $A^* = A_1^* \circ \dots \circ A_{|\mathcal{V}|}^*$ for a composition $A = A_{|\mathcal{V}|} \circ \dots \circ A_1$. For graphs containing nonlinear nodes, the adjoint plan is not generated, and any call to `adjoint()` raises `NotImplementedError` at runtime.

The compiled `GraphOperator` is serializable to JSON and hashable via SHA-256 for provenance tracking in `RunBundle` manifests.

Template library. The `graph_templates.yaml` registry contains 170 templates spanning all five carrier families. Of these, 12 modalities carry full end-to-end Scenario I–IV correction validation across all five carrier families (Supplementary Table S3). Representative modalities by carrier family include:

- **Optical and X-ray photons:** CASSI, SPC, CACTI, structured illumination microscopy (SIM), confocal, light-sheet, holography, ptychography, Fourier ptychographic

749 microscopy (FPM), optical coherence tomography (OCT), lensless imaging, light field,
 750 integral imaging, neural radiance fields (NeRF), Gaussian splatting, fluorescence life-
 751 time imaging (FLIM), diffuse optical tomography (DOT), phase retrieval, X-ray com-
 752 puted tomography (CT), and cone-beam CT (CBCT).

753 • **Electrons:** Electron diffraction, electron backscatter diffraction (EBSD), electron
 754 energy loss spectroscopy (EELS), and electron holography.

755 • **Nuclear spins (MRI):** Functional MRI (fMRI), diffusion-weighted MRI (DW-MRI),
 756 and magnetic resonance spectroscopy (MRS).

757 • **Acoustic waves:** Ultrasound B-mode, Doppler ultrasound, shear-wave elastogra-
 758 phy, sonar, and photoacoustic tomography (combines optical excitation with acoustic
 759 detection).

760 • **Particle-beam probes:** Neutron tomography, proton CT, and muon tomography.
 761 Each decomposes into existing primitives with no extension required: neutron CT
 762 follows the X-ray CT DAG ($P \rightarrow \Lambda \rightarrow D$, Beer–Lambert attenuation with nuclear cross-
 763 section μ_n); proton CT inserts a Bethe–Bloch stopping-power stage as Transform Λ
 764 ($\#11$); muon tomography uses Scatter R ($\#10$) for multiple Coulomb scattering and
 765 POCA vertex reconstruction (Supplementary Note S24).

766 **Fidelity levels.** Each template is parameterized by a fidelity level that controls the degree
 767 of physical realism in the simulated forward model:

768 **Level 1 (Linear, shift-invariant):** The forward model is a linear, spatially uniform operator—
 769 the simplest approximation, suitable for initial diagnostics and rapid prototyping.

770 **Level 2 (Linear, shift-variant):** Spatially varying operator parameters (e.g. non-uniform
 771 illumination, position-dependent PSF, multi-coil sensitivity maps in MRI). Adds a
 772 modality-appropriate noise model (Poisson shot noise plus Gaussian read noise for
 773 photon-counting modalities, Rician noise for MRI, Poisson for CT).

774 **Level 3 (Nonlinear, ray/wave-based):** Includes nonlinear effects such as beam hard-
 775 ening, phase wrapping, and scattering, captured by the Transform Λ primitive. Per-
 776 turbation families and ranges are specified in `mismatch_db.yaml`.

777 **Level 4 (Full-wave / Monte Carlo):** Complete physical simulation including wave-optical
 778 propagation, spatially varying aberrations, detector nonlinearities, and environmental
 779 drift. Currently implemented for holography and ptychography.

780 All 11 primitives in the library $\mathcal{B}$ apply uniformly across every fidelity level; the levels
 781 organize simulation complexity, not theorem scope.

Triad Decomposition Formalization

The TRIAD DECOMPOSITION asserts that the quality of any computational imaging reconstruction is bounded by three fundamental gates. Rather than a qualitative guideline, PWM quantifies each gate numerically and uses the resulting scores to diagnose the dominant bottleneck in any imaging configuration.

Gate 1 (Recoverability). Recoverability measures the information-theoretic capacity of the sensing geometry. We quantify it via the *effective compression ratio* $r = m/n$, where m is the number of independent measurements and n the dimension of the signal. The `compression_db.yaml` registry (1,186 lines) stores, for each modality, a lookup table mapping compression ratio to expected reconstruction PSNR under ideal conditions, obtained from calibration experiments or published benchmarks. Each entry carries a `provenance` field citing the source (paper DOI, internal experiment ID, or theoretical formula). Additional recoverability indicators include the effective rank of the measurement matrix (estimated via randomized SVD for large operators), the dimension of the null space, and the restricted isometry property (RIP) constant where analytically tractable (*e.g.*, for Gaussian random projections in SPC).

Gate 2 (Carrier Budget). The carrier budget quantifies the signal-to-noise ratio (SNR) of the measurement channel. The photon-budget module consumes the `photon_db.yaml` registry which stores, per modality, a deterministic photon model parameterized by source power, quantum efficiency, exposure time, and detector characteristics. It classifies the noise regime into one of three categories: *shot-limited* (Poisson-dominated, $\text{SNR} \propto \sqrt{N_{\text{photon}}}$), *read-limited* (Gaussian read noise dominates, $\text{SNR} \propto N_{\text{photon}}/\sigma_{\text{read}}$), and *dark-current-limited* (long exposures where dark current accumulation dominates). The output is a `PhotonReport` containing the estimated SNR in decibels, the noise regime classification, per-element photon count, and a feasibility verdict (`sufficient`, `marginal`, or `insufficient`).

Gate 3 (Operator Mismatch). Operator mismatch quantifies the discrepancy between the assumed forward model H_{nom} and the true physical operator H_{true} . The mismatch scoring module consults `mismatch_db.yaml` which catalogs, for each modality, the set of mismatch parameters (spatial shifts, rotational offsets, dispersion errors, PSF deviations, coil sensitivity errors, center-of-rotation offsets, *etc.*), their typical ranges, and available correction methods. The mismatch severity score $s \in [0, 1]$ is computed as the normalized ℓ_2 distance $\|\theta_{\text{true}} - \theta_{\text{nom}}\|/\|\theta_{\text{range}}\|$, where θ_{range} is the per-parameter dynamic range from the registry. Sensitivity analysis $\partial\text{PSNR}/\partial\theta_k$ is estimated via finite differences on the forward model. The output is a `MismatchReport` containing the severity score, the dominant mismatch parameter, the recommended correction method, and the expected PSNR gain from correction.

818 **Gate binding determination.** Given reconstruction results under the four-scenario pro-
 819 tocol (the Evaluation Protocol section below), PWM identifies the dominant gate by com-
 820 paring three cost terms:

$$C_{\text{mismatch}} = \text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}} \quad (2)$$

$$C_{\text{noise}} = \text{PSNR}_{\text{ideal}} - \text{PSNR}_{\text{noisy}} \quad (3)$$

$$C_{\text{recover}} = \text{PSNR}_{\text{limit}} - \text{PSNR}_{\text{I}} \quad (4)$$

821 where PSNR_{I} is the reconstruction PSNR under Scenario I (ideal operator), PSNR_{II} under
 822 Scenario II (mismatched operator), $\text{PSNR}_{\text{noisy}}$ under the corresponding noisy condition,
 823 and $\text{PSNR}_{\text{limit}}$ is the theoretical upper bound from the compression table. The dominant
 824 gate is $\arg \max_g C_g$.

825 **TriadReport.** For every diagnosis, PWM produces a TRIADREPORT: a Pydantic-validated
 826 structured artifact comprising `dominant_gate` (enum: `recoverability`, `carrier_budget`,
 827 `operator_mismatch`), `evidence_scores` (three floats, one per gate), `confidence_interval`
 828 (float, 95% CI width from bootstrap), `recommended_action` (string, *e.g.* “increase compres-
 829 sion ratio” or “apply mismatch correction”), and `parameter_sensitivities` (dictionary
 830 mapping each mismatch parameter name to its $\partial \text{PSNR} / \partial \theta_k$ value). The TRIADREPORT is
 831 mandatory—PWM does not permit a reconstruction to be reported without an accompany-
 832 ing diagnosis.

833 **Recovery ratio.** We define the *recovery ratio*

$$\rho = \frac{\text{PSNR}_{\text{IV}} - \text{PSNR}_{\text{II}}}{\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}}} \quad (5)$$

834 which lies in $[0, 1]$ under standard convexity conditions (see Supplementary Note 1 for
 835 formal analysis; values $\rho > 1$ are possible when the corrected operator provides beneficial
 836 regularization). $\rho = 0$ indicates that calibration yields no benefit (mismatch is not the
 837 bottleneck), while $\rho = 1$ indicates that calibration fully closes the mismatch gap.

838 Agent System Architecture

839 The three gates are implemented as a deterministic scoring pipeline: an orchestrator maps
 840 the user request to a modality from the registry, then dispatches three parallel scoring
 841 modules—one per gate—whose reports are fused by a bottleneck classifier that identifies
 842 the dominant gate and by a negotiator that enforces cross-gate veto rules (e.g., rejecting
 843 reconstruction when photon starvation coincides with aggressive compression). All quanti-
 844 tative decisions are deterministic; an LLM is used only as an optional fallback for natural-
 845 language modality mapping and is never on the scoring path. The full agent architecture

846 and ablation studies are described in a companion paper.

847 **Contract system.** Inter-agent communication uses 25 Pydantic v2 contract models. All
 848 contracts inherit from `StrictBaseModel`, which enforces `extra="forbid"` (no unexpected
 849 fields), `validate_assignment=True` (mutations re-validated), and a model validator that
 850 rejects NaN and Inf in any float field. Bounded scores use `Field(ge=0.0, le=1.0)`. Enums
 851 are string enums for human-readable JSON serialization. This design ensures that pipeline
 852 failures surface immediately as validation errors rather than propagating silently.

853 **YAML registries.** The system is driven by 9 YAML registries totalling 7,034 lines:
 854 `modalities.yaml` (modality definitions), `graph_templates.yaml` (OperatorGraph skele-
 855 tons), `photon_db.yaml` (photon models), `mismatch_db.yaml` (mismatch parameters and
 856 correction methods), `compression_db.yaml` (recoverability tables with provenance), `solver_registry.yaml`
 857 (solver configurations), `primitives.yaml` (primitive operator metadata), `dataset_registry.yaml`
 858 (dataset locations and formats), and `acceptance_thresholds.yaml` (pass/fail thresholds
 859 per metric).

860 Correction Algorithms

861 We implement two complementary algorithms for operator mismatch correction. Crucially,
 862 both algorithms operate on the forward operator parameters θ rather than the reconstruc-
 863 tion solver weights, making them *solver-agnostic*: the corrected operator $H(\hat{\theta})$ benefits any
 864 downstream solver (GAP-TV, MST-L, HDNet³⁹, CST, *etc.*) without retraining.

865 **Algorithm 1: Hierarchical Beam Search.** The coarse correction phase employs a
 866 hierarchical search strategy to rapidly explore the mismatch parameter space. For CASSI,
 867 the five-parameter mismatch model comprises mask affine parameters (spatial shifts dx , dy
 868 and rotation θ) and dispersion parameters (slope a_1 and axis angle α); an optional sixth
 869 parameter, PSF width σ_{psf} , is available but not used in the primary experiments. The
 870 algorithm proceeds as follows:

- 871 1. **1D sweeps.** Each parameter is swept independently over its full range while holding
 872 others at nominal values. This produces five 1D cost curves from which coarse optima
 873 are extracted.
- 874 2. **3D beam search.** The mask affine subspace (dx, dy, θ) is searched over a $5 \times 5 \times 5$
 875 grid centered on the 1D optima. The top- k ($k = 5$) candidates by reconstruction
 876 PSNR are retained.
- 877 3. **2D beam search.** For each retained mask candidate, the dispersion subspace (a_1, α)
 878 is searched over a 5×7 grid. The joint top- k candidates are retained.

879 **4. Coordinate descent refinement.** Three rounds of univariate refinement on each
880 parameter, shrinking the search interval by factor 2 at each round, produce the final
881 estimate $\hat{\theta}_{\text{Alg1}}$.

882 Total runtime is approximately 300 seconds per scene on a single GPU. Accuracy is
883 ± 0.1 – 0.2 pixels for spatial parameters and $\pm 0.05^\circ$ for angular parameters.

884 **Algorithm 2: Joint Gradient Refinement.** The fine correction phase uses a differen-
885 tiable forward model to jointly optimize all mismatch parameters via gradient descent. The
886 key components are:

- 887 **1. Differentiable mask warp.** The binary mask is warped by a continuous affine
888 transformation using bilinear interpolation, implemented as a custom PyTorch module
889 (`DifferentiableMaskWarpFixed`). The mask values are passed through a straight-
890 through estimator (STE) to maintain binary structure while permitting gradient flow.
- 891 **2. Differentiable forward model.** The CASSI forward model $y = \text{CASSI}(x; \theta)$ is
892 implemented as a differentiable PyTorch module (`DifferentiableCassiForwardSTE`)
893 that accepts mismatch parameters as differentiable inputs.
- 894 **3. GPU grid initialization.** A full-range 3D grid search over (dx, dy, θ) with $9 \times 9 \times 7 =$
895 567 points provides diverse starting candidates. The top 9 candidates seed multi-start
896 gradient refinement.
- 897 **4. Staged gradient refinement.** Each of the 9 candidates is refined using Adam
898 optimization (learning rate 10^{-2} , decaying to 10^{-3}) for 200 steps. For each candidate,
899 4 random restarts with jittered initialization guard against local minima. The loss
900 function is the negative PSNR computed via an unrolled K -iteration differentiable
901 GAP-TV solver (`DifferentiableGAPTV`, $K = 10$ unrolled iterations).

902 Total runtime for Algorithm 2 is approximately 3,200 seconds ($200 \text{ steps} \times 4 \text{ restarts} \times$
903 $9 \text{ candidates with early stopping}$). Accuracy improves to ± 0.05 – 0.1 pixels, a 3 – $5\times$ improve-
904 ment over Algorithm 1. The two algorithms are used sequentially in practice: Algorithm 1
905 provides a warm start, and Algorithm 2 refines to sub-pixel precision.

906 Evaluation Protocol

907 **Four-Scenario Protocol.** We evaluate every modality under four standardized scenarios
908 that isolate different sources of quality degradation:

909 **Scenario I (Ideal):** $y_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with H_{true} . In this scenario the system
910 is perfectly calibrated ($H_{\text{true}} = H_{\text{nom}}$), so the operator used for reconstruction matches
911 the one that generated the data. This yields the oracle upper bound on reconstruction
912 quality, limited only by the sensing geometry and solver convergence.

913 **Scenario II (Mismatch):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with H_{nom} ($H_{\text{nom}} \neq H_{\text{true}}$). This
 914 is the standard operating condition in practice: the measurement is generated by the
 915 true physics, but the reconstruction uses a nominal (potentially mismatched) forward
 916 model.

917 **Scenario III (Oracle):** Same measurements as Scenario II ($\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$ with $H_{\text{true}} \neq$
 918 H_{nom}); reconstruct with H_{true} instead of H_{nom} . Provides the correction ceiling: the
 919 best reconstruction achievable when the true operator is known exactly, applied to
 920 data that were sensed by the mismatched system. The gap between Scenario III and
 921 Scenario I reveals the irreducible loss from the degraded sensing configuration itself
 922 (e.g., a shifted mask pattern is suboptimal even when perfectly known).

923 **Scenario IV (Corrected):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with $\hat{H} = H(\hat{\theta})$ where $\hat{\theta}$ is
 924 estimated by Algorithms 1 and 2. This quantifies the benefit of mismatch calibration.

925 **Metrics.** Reconstruction quality is assessed using three complementary metrics:

- 926 • **PSNR** (peak signal-to-noise ratio, in dB): the primary metric, computed per scene
 927 and averaged. For signals normalized to $[0, 1]$, $\text{PSNR} = 10 \log_{10}(1/\text{MSE})$. For SPC
 928 data normalized to $[0, 255]$, the peak value is 255.
- 929 • **SSIM** (structural similarity index): captures perceptual quality including luminance,
 930 contrast, and structural components, computed with a Gaussian window of width 11
 931 and standard deviation 1.5.
- 932 • **SAM** (spectral angle mapper): for hyperspectral modalities (CASSI), measures the
 933 angle between predicted and true spectral vectors at each spatial location, reported
 934 in degrees. Lower is better.

935 **Datasets.**

- 936 • **CASSI:** 10 scenes from the KAIST dataset³⁶, each a $256 \times 256 \times 28$ spectral cube
 937 (28 spectral bands from 450 nm to 650 nm). Data range $[0, 1]$.
- 938 • **CACTI:** 6 benchmark videos, each $256 \times 256 \times 8$ (8 temporal frames encoded per
 939 snapshot). Data range $[0, 1]$.
- 940 • **SPC:** 11 natural images from the Set11 benchmark, each 256×256 grayscale. Data
 941 range $[0, 255]$.

942 All per-scene metrics are reported individually as well as averaged, and all reconstruction
 943 arrays are saved as NumPy NPZ files.

Experimental Details

Hardware. All experiments are conducted on a single NVIDIA GPU. Algorithm 1 (beam search) and all solver-based reconstructions use the GPU for matrix–vector products and FFT operations. Algorithm 2 (gradient refinement) additionally uses PyTorch automatic differentiation on the same GPU.

CASSI configuration. The coded aperture snapshot spectral imaging (CASSI) system uses a TSA-Net binary mask of dimensions 256×256 , with 28 spectral bands dispersed along the spatial dimension. The five-parameter mismatch model $\theta = (dx, dy, \theta, a_1, \alpha)$ describes: mask spatial shift in x (dx , pixels), mask spatial shift in y (dy , pixels), mask rotation angle (θ , degrees), dispersion slope (a_1 , pixels per band), and dispersion axis angle (α , degrees). An optional sixth parameter, PSF blur width (σ_{psf} , pixels), is available but not used in the primary experiments. These mismatch parameter values were determined through systematic characterization of realistic CASSI assembly errors (Supplementary Note 8). The true mismatch parameters are $\theta_{\text{true}} = (dx = 0.5 \text{ px}, dy = 0.3 \text{ px}, \theta = 0.1^\circ, a_1 = 2.02, \alpha = 0.15^\circ)$. Solvers evaluated include TwIST⁴⁰, GAP-TV⁴¹, DGSMP⁴², MST-L⁷, and CST-L⁴³, all of which receive the same operator and differ only in their reconstruction algorithm. The supplementary per-scene analysis additionally includes DeSCI⁴⁴ and HDNet³⁹. *Real-data validation.* The TSA real hyperspectral dataset¹² consists of 5 scenes at 660×660 spatial resolution with 28 spectral bands and mask-shift step 2. Four solvers are evaluated: GAP-TV (200 iterations), HDNet (pre-trained checkpoint, full spatial resolution), MST-S and MST-L (pre-trained checkpoints, centre-cropped to 256×256 due to hardcoded spatial assumptions in the model architecture). The coded aperture mask is perturbed by $dx = 0.5 \text{ px}$, $dy = 0.3 \text{ px}$ to simulate assembly-induced mismatch. No ground truth is available; quality is assessed via the normalised measurement residual $r = \|\mathbf{y} - H\hat{\mathbf{x}}\|^2 / \|\mathbf{y}\|^2$.

CACTI configuration. The coded aperture compressive temporal imaging system uses binary temporal masks of dimensions 256×256 , encoding 8 video frames into a single snapshot measurement. Mismatch is parameterized as a temporal mask timing offset (sub-frame shift). The default solver is EfficientSCI²¹. *Real-data validation.* The EfficientSCI real temporal dataset²¹ consists of 4 dynamic scenes (duomino, hand, pendulumBall, waterBalloon) at 512×512 with compression ratio 10. The real mask is stored separately from the measurement data. Two solvers are evaluated: GAP-TV (50 iterations) and PnP-FFDNet (50 iterations with FFDNet denoiser). Mismatch is induced by shifting the mask by $dx = 0.5 \text{ px}$, $dy = 0.3 \text{ px}$. Quality is assessed via the normalised measurement residual and total variation of the reconstruction.

SPC configuration. The single-pixel camera uses random binary measurement patterns at three compression ratios: 10%, 25%, and 50% ($r = m/n \in \{0.10, 0.25, 0.50\}$). Mismatch

is modeled as an exponential gain drift ($g_i = \exp(-\alpha \cdot i)$) on the measurement matrix. The default solver is FISTA-TV with total-variation regularization.

MRI configuration. Cartesian k -space sampling with $4\times$ acceleration (25% of k -space lines acquired). Mismatch is parameterized as a 5% multiplicative error in the coil sensitivity maps used for parallel imaging reconstruction. The default solver is CG-SENSE¹⁷ with ℓ_1 -wavelet regularization ($\lambda = 10^{-3}$, 30 iterations). *Scenario IV calibration (ESPIRiT-adopted).* For MRI, Scenario IV adopts ESPIRiT¹⁰ as the calibration method: coil sensitivity maps are estimated directly from the central k -space autocalibration signal (ACS) region via eigenvalue decomposition of the calibration matrix, then used in the CG-SENSE reconstruction. When ACS data is sufficient (≥ 48 lines), ESPIRiT achieves 85–95% of the oracle correction ceiling. For data-limited regimes (< 32 ACS lines) where ESPIRiT degrades, the fallback is a grid search over per-coil amplitude and phase scaling (beam-search); this fallback is also the Sc. IV method for all modalities without an established specialist calibration algorithm (CASSI, CACTI, compressive holography, ultrasound). The protocol evaluates three conditions—Sc. I (true maps), Sc. II (5% mismatched maps), and Sc. IV (ESPIRiT-calibrated maps)—on the same CG-SENSE solver to isolate calibration quality from solver design.

CT configuration. Fan-beam geometry with 180 projections over 180° . Mismatch is modeled as a center-of-rotation (CoR) offset, which produces characteristic arc artifacts in the reconstruction. The default solver is filtered back-projection (FBP)¹⁵ with a Ram-Lak filter, supplemented by iterative SART for comparison.

Controlled hardware experiment protocol. The software-perturbation protocol above applies calibrated mask shifts to existing real measurements. A full hardware-in-the-loop validation requires physically displacing the coded aperture mask and re-acquiring data. The protocol proceeds as follows: (i) acquire a baseline dataset with the mask at its factory-calibrated position; (ii) physically translate the mask by a known displacement ($\Delta x \in \{0.25, 0.5, 1.0\}$ px equivalent, verified by micrometer stage) and re-acquire under identical illumination; (iii) reconstruct both datasets with the factory mask specification and compute the PSNR degradation and measurement residual; (iv) apply PWM autonomous calibration and measure recovery. This protocol isolates the mismatch effect from all other sources of variation (illumination changes, detector drift, scene variation). Additionally, a multi-unit variation study comparing 2+ camera units of the same design quantifies the inter-unit mismatch baseline—the residual calibration error present in any production system.

Clinical CT phantom configuration. For clinical translation, PWM is evaluated on CT quality assurance using the ACR CT accreditation phantom (Gammex 464). The phantom

contains inserts of known attenuation (bone ~ 955 HU, air ~ -1000 HU, acrylic ~ 121 HU, polyethylene ~ -96 HU) and geometric targets for measuring spatial resolution, slice thickness, and low-contrast detectability. Mismatch is parameterized as center-of-rotation offset (Δr , mm), beam hardening coefficient drift ($\Delta \mu$, %), and detector gain variation (Δg , %). Ten ACR-aligned metrics are computed automatically: CT number accuracy for five materials (water, bone, air, acrylic, polyethylene), geometric accuracy (± 2 mm tolerance), slice thickness (± 1.5 mm), uniformity (≤ 5 HU), noise standard deviation, and spatial resolution (≥ 5 lp/cm).

Clinical MRI validation configuration. For MRI clinical validation, PWM processes multi-coil k -space data from public datasets (fastMRI⁴⁵). Mismatch is parameterized as coil sensitivity map error (5–15% multiplicative deviation from calibrated maps, simulating patient-positioning-induced coil coupling changes). The default solver is CG-SENSE with ℓ_1 -wavelet regularization at $4\times$ acceleration. Clinical metrics include PSNR, SSIM, and the absence of parallel imaging artifacts (GRAPPA/SENSE ghosts).

Ultrasound configuration. Pulse-echo B-mode imaging is simulated with a 64-element linear array, 5 MHz center frequency, element pitch 0.3 mm, sampling rate 100 MHz with 1024 time samples, and speed of sound (SoS) $c = 1540$ m/s. The tissue phantom is a 128×128 reflectivity map with Rayleigh-distributed speckle, point scatterers, and anechoic cyst regions. Frequency-dependent attenuation follows $\alpha(f) = 0.5 \text{ dB cm}^{-1} \text{ MHz}^{-1}$. Mismatch is parameterized as SoS error $\Delta c \in \{50, 100, 200, 400\}$ m/s, perturbing time-of-flight calculations. The default solver is delay-and-sum (DAS) beamforming (operator adjoint). Calibration uses grid search over 21 SoS candidates in $[1440, 1640]$ m/s, selecting the value that maximizes reconstruction PSNR.

Cryo-EM configuration. Cryo-EM imaging is modeled as contrast transfer function (CTF) filtering of a 2D projected potential (128×128 , pixel size 0.1 nm). The CTF is parameterized by defocus $\Delta f = -2000$ nm, spherical aberration $C_s = 2.0$ mm, electron wavelength $\lambda = 0.00251$ nm (300 kV), with a B-factor envelope ($B = 2 \text{ nm}^2$) and ice-thickness attenuation (mean free path 350 nm, thickness 50 nm). Mismatch is parameterized as defocus error $\Delta(\Delta f) \in \{100, 200, 500, 1000\}$ nm. The default solver is Wiener filtering with $\text{SNR} = 50$. Calibration uses grid search over 51 defocus values in $[-3000, -500]$ nm, selecting the value that maximizes reconstruction PSNR.

CT detector offset configuration. Parallel-beam CT is simulated with 360 projection angles over $[0, \pi)$ and 128 detector elements. The object is a 128×128 Shepp–Logan phantom. Detector offset mismatch is simulated by shifting the sinogram along the detector axis by $\Delta s \in \{2, 5, 10, 20\}$ pixels. The default solver is filtered backprojection (FBP) with

1051 Shepp–Logan filter. Calibration uses grid search over 21 offset candidates in $[-25, 25]$ px,
 1052 selecting the value that maximizes reconstruction PSNR.

1053 **Compressive holography configuration.** Multi-depth inline holography encodes a $(4 \times$
 1054 $64 \times 64)$ 3D object into a single (64×64) hologram via Fresnel propagation at four depth
 1055 planes with spacing $200 \mu\text{m}$, wavelength $\lambda = 532 \text{ nm}$, and pixel pitch $5 \mu\text{m}$. Mismatch is
 1056 parameterized as propagation distance error $\Delta z \in \{10, 50, 100, 200\} \mu\text{m}$ applied uniformly to
 1057 all depth planes. The default solver is FISTA-TV (80 iterations, $\lambda_{\text{TV}} = 0.005$). Calibration
 1058 uses hologram residual minimisation: grid search over distance error candidates with the
 1059 search range clamped to ensure all propagation distances remain positive.

1060 **Fluorescence microscopy configuration.** Widefield fluorescence imaging uses a dual-
 1061 PSF Stokes-shift model with Gaussian excitation PSF ($\sigma_{\text{ex}} = 1.5 \text{ px}$) and emission PSF
 1062 ($\sigma_{\text{em}} = 2.0 \text{ px}$), quantum yield $\eta = 0.7$, and background level $b = 0.02$. The specimen
 1063 is a 64×64 image with fluorescent puncta, diffuse organelles, and filamentary structures.
 1064 Mismatch is parameterized as PSF sigma error $\Delta\sigma \in \{0.1, 0.3, 0.5, 1.0\} \text{ px}$ applied to both
 1065 excitation and emission PSFs. The default solver is Richardson–Lucy (80 iterations). Cal-
 1066 ibration uses a 9×9 grid search over $(\sigma_{\text{ex}}, \sigma_{\text{em}})$ centered at the nominal values, selecting
 1067 the pair that minimizes the measurement residual.

1068 **Real experimental data.** Three of the five multi-phantom modalities are validated on
 1069 real experimental data from established benchmarks: ultrasound uses PICMUS experimen-
 1070 tal phantom recordings and in-vivo carotid scans²⁷ plus one DeepUS CIRS-040GSE acqui-
 1071 sition²⁸; cryo-EM uses five EMDB 3D density maps (EMD-5778, EMD-2984, EMD-6287,
 1072 EMD-11103, EMD-21375) projected at random orientations; CT uses three LoDoPaB-CT
 1073 patient slices²⁴, one FIPS walnut μCT scan²⁵, and one HTC 2022 sample²⁶. Compressive
 1074 holography and fluorescence microscopy retain synthetic multi-phantom benchmarks as no
 1075 public datasets with calibrated ground truth are available for these modalities.

1076 Statistical Analysis

1077 **Per-scene reporting.** All metrics are reported per scene, not merely as dataset averages.
 1078 This enables identification of scene-dependent failure modes (*e.g.*, spectrally flat scenes that
 1079 are inherently harder for CASSI, or textureless regions that challenge SPC).

1080 **Summary statistics.** For each modality and scenario, we report the mean $\pm$ standard
 1081 deviation of PSNR, SSIM, and SAM across all scenes. For CASSI (10 scenes), we addition-
 1082 ally report the per-band PSNR to assess spectral uniformity of reconstruction quality.

Recovery ratio confidence intervals. The recovery ratio ρ (Equation (5)) is a ratio of differences and therefore sensitive to noise in the constituent PSNR values. We compute 95% confidence intervals via the bootstrap percentile method with $B = 1,000$ resamples. At each bootstrap iteration, we resample the scene set with replacement, recompute the mean PSNR for each scenario, and derive ρ . The 2.5th and 97.5th percentiles of the bootstrap distribution define the 95% CI.

Parameter recovery accuracy. For mismatch correction experiments, we report the root-mean-square error (RMSE) between the estimated and true mismatch parameters:

$$\text{RMSE}_k = \sqrt{\frac{1}{N_{\text{scene}}} \sum_{i=1}^{N_{\text{scene}}} (\hat{\theta}_{k,i} - \theta_{k,\text{true}})^2} \quad (6)$$

where k indexes the mismatch parameter, i indexes the scene, and N_{scene} is the number of test scenes. Uncertainty in the RMSE is estimated via bootstrap ($B = 1,000$).

Ablation significance. Ablation studies (removal of the photon-budget, recoverability, or mismatch scoring module, or RunBundle discipline) are evaluated by comparing the full-pipeline PSNR against each ablated variant. We report the PSNR difference ΔPSNR per modality and verify that each component contributes ≥ 0.5 dB across all validated modalities, establishing practical significance.

Code and Data Availability

Source code. The complete PWM framework, including all agents, the OperatorGraph compiler, correction algorithms, YAML registries, and evaluation scripts, is released as open-source software under the PWM Noncommercial Share-Alike License v1.0 at https://github.com/integritynoble/Physics_World_Model. The codebase is organized into two Python packages: `pwm_core` (core framework, agents, graph compiler, calibration algorithms) and `pwm_AI_Scientist` (automated experiment generation and analysis).

Reconstruction data. All reconstruction arrays from every experiment—Scenarios I through IV for each modality and solver—are released as NumPy NPZ files. Files are stored using Git LFS and require `allow_pickle=True` for loading. Data ranges are standardized: CASSI and CACTI reconstructions are normalized to $[0, 1]$; SPC reconstructions are in $[0, 255]$.

Experiment manifests. Every experiment is recorded in a RunBundle v0.3.0 manifest containing: the git commit hash at execution time, all random number generator seeds, platform information (Python version, GPU model, CUDA version), SHA-256 hashes of all

1113 input data and output artifacts, metric values, and wall-clock timestamps. These manifests
1114 enable exact reproduction of every reported result.

1115 **Registry data.** All 9 YAML registries (7,034 lines total) that drive the agent system—
1116 including modality definitions, graph templates, photon models, mismatch databases, com-
1117 pression tables, solver configurations, primitive specifications, dataset paths, and acceptance
1118 thresholds—are publicly available in the repository under `packages/pwm_core/contrib/`.
1119 The `ExperimentSpec` JSON schemas used for pipeline input validation are included along-
1120 side worked examples in `examples/`.

1121 References

- 1122 [1] Chengshuai Yang. The finite primitive basis theorem for computational imaging: For-
1123 mal foundations of the OperatorGraph representation, 2026. URL [https://arxiv.](https://arxiv.org/abs/2602.20550)
1124 [org/abs/2602.20550](https://arxiv.org/abs/2602.20550).
- 1125 [2] Emmanuel J. Candès and Michael B. Wakin. An introduction to compressive sampling.
1126 *IEEE Signal Processing Magazine*, 25(2):21–30, 2008. doi: 10.1109/MSP.2007.914731.
- 1127 [3] David L. Donoho. Compressed sensing. *IEEE Transactions on Information Theory*,
1128 52(4):1289–1306, 2006. doi: 10.1109/TIT.2006.871582.
- 1129 [4] Singanallur V. Venkatakrishnan, Charles A. Bouman, and Brendt Wohlberg. Plug-
1130 and-play priors for model based reconstruction. In *Proceedings of the IEEE Global*
1131 *Conference on Signal and Information Processing (GlobalSIP)*, pages 945–948, 2013.
1132 doi: 10.1109/GlobalSIP.2013.6737048.
- 1133 [5] Xin Yuan, Yang Liu, Jinli Suo, and Qionghai Dai. Plug-and-play algorithms for video
1134 snapshot compressive imaging. *IEEE Transactions on Pattern Analysis and Machine*
1135 *Intelligence*, 44(10):7093–7111, 2021.
- 1136 [6] Vishal Monga, Yuelong Li, and Yonina C. Eldar. Algorithm unrolling: Interpretable,
1137 efficient deep learning for signal and image processing. *IEEE Signal Processing Maga-*
1138 *zine*, 38(2):18–44, 2021. doi: 10.1109/MSP.2020.3016905.
- 1139 [7] Yuanhao Cai, Jing Lin, Xiaowan Hu, Haoqian Wang, Xin Yuan, Yulun Zhang, Radu
1140 Timofte, and Luc Van Gool. Mask-guided spectral-wise transformer for efficient hy-
1141 perspectral image reconstruction. In *Proceedings of the IEEE/CVF Conference on*
1142 *Computer Vision and Pattern Recognition (CVPR)*, pages 17502–17511, 2022.
- 1143 [8] Xin Yuan, David J. Brady, and Aggelos K. Katsaggelos. Snapshot compressive imaging:
1144 Theory, algorithms, and applications. *IEEE Signal Processing Magazine*, 38(2):65–88,
1145 2021.

- 1146 [9] Vegard Antun, Francesco Renna, Clarice Poon, Ben Adcock, and Anders C. Hansen.
1147 On instabilities of deep learning in image reconstruction and the potential costs of
1148 AI. *Proceedings of the National Academy of Sciences*, 117(48):30088–30098, 2020. doi:
1149 10.1073/pnas.1907377117.
- 1150 [10] Martin Uecker, Peng Lai, Mark J. Murphy, Patrick Virtue, Michael Elad, John M.
1151 Pauly, Shreyas S. Vasanawala, and Michael Lustig. ESPIRiT — an eigenvalue approach
1152 to autocalibrating parallel MRI: where SENSE meets GRAPPA. *Magnetic Resonance*
1153 *in Medicine*, 71(3):990–1001, 2014. doi: 10.1002/mrm.24751.
- 1154 [11] Andrew M. Maiden and John M. Rodenburg. An improved ptychographical phase
1155 retrieval algorithm for diffractive imaging. *Ultramicroscopy*, 109(10):1256–1262, 2009.
1156 doi: 10.1016/j.ultramic.2009.05.012.
- 1157 [12] Ashwin A. Wagadarikar, Renu John, Rebecca Willett, and David J. Brady. Single
1158 disperser design for coded aperture snapshot spectral imaging. *Applied Optics*, 47(10):
1159 B44–B51, 2008. doi: 10.1364/AO.47.000B44.
- 1160 [13] Patrick Llull, Xuejun Liao, Xin Yuan, Jianbo Yang, David Kittle, Lawrence Carin,
1161 Guillermo Sapiro, and David J. Brady. Coded aperture compressive temporal imaging.
1162 *Optics Express*, 21(9):10526–10545, 2013. doi: 10.1364/OE.21.010526.
- 1163 [14] Marco F. Duarte, Mark A. Davenport, Dharmpal Takhar, Jason N. Laska, Ting Sun,
1164 Kevin F. Kelly, and Richard G. Baraniuk. Single-pixel imaging via compressive sam-
1165 pling. *IEEE Signal Processing Magazine*, 25(2):83–91, 2008. doi: 10.1109/MSP.2007.
1166 914730.
- 1167 [15] L. A. Feldkamp, L. C. Davis, and J. W. Kress. Practical cone-beam algorithm. *Jour-*
1168 *nal of the Optical Society of America A*, 1(6):612–619, 1984. doi: 10.1364/JOSAA.1.
1169 000612.
- 1170 [16] J. M. Rodenburg and H. M. L. Faulkner. A phase retrieval algorithm for shifting
1171 illumination. *Applied Physics Letters*, 85(20):4795–4797, 2004. doi: 10.1063/1.1823034.
- 1172 [17] Klaas P. Pruessmann, Markus Weiger, Markus B. Scheidegger, and Peter Boesiger.
1173 SENSE: Sensitivity encoding for fast MRI. *Magnetic Resonance in Medicine*, 42(5):
1174 952–962, 1999. doi: 10.1002/(SICI)1522-2594(199911)42:5<952::AID-MRM16>3.0.CO;
1175 2-S.
- 1176 [18] Michael Lustig, David Donoho, and John M. Pauly. Sparse MRI: The application of
1177 compressed sensing for rapid MR imaging. *Magnetic Resonance in Medicine*, 58(6):
1178 1182–1195, 2007. doi: 10.1002/mrm.21391.
- 1179 [19] David J. Brady. *Optical Imaging and Spectroscopy*. John Wiley & Sons, 2009.

- [20] Chengshuai Yang. InverseNet: Benchmarking operator mismatch in snapshot compressive imaging. *arXiv preprint arXiv:2603.04538*, 2026.
- [21] Lishun Wang, Miao Cao, and Xin Yuan. EfficientSCI: Densely connected network with space-time factorization for large-scale video snapshot compressive imaging. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 18477–18486, 2023.
- [22] Nick Antipa, Grace Kuo, Reinhard Heckel, Ben Mildenhall, Emrah Bostan, Ren Ng, and Laura Waller. DiffuserCam: lensless single-exposure 3D imaging. *Optica*, 5(1):1–9, 2018. doi: 10.1364/OPTICA.5.000001. DiffuserCam DLMD dataset: HuggingFace bezzam/DiffuserCam-Lensless-Mirflickr-Dataset.
- [23] David J. Brady, Kerkil Choi, Daniel L. Marks, Ryoichi Horisaki, and Sehoon Lim. Compressive holography. *Optics Express*, 17(15):13040–13049, 2009.
- [24] Johannes Leuschner, Maximilian Schmidt, Daniel Otero Baguer, and Peter Maaß. LoDoPaB-CT, a benchmark dataset for low-dose computed tomography reconstruction. *Scientific Data*, 8:109, 2021. doi: 10.1038/s41597-021-00893-z.
- [25] Keijo Hämäläinen, Lauri Harhanen, Andreas Hauptmann, Aki Kallonen, Esa Niemi, and Samuli Siltanen. Total variation regularization for large-scale X-ray tomography. *International Journal of Tomography and Simulation*, 28(1):1–25, 2015. FIPS walnut dataset: Zenodo 1254206.
- [26] Alexander Meaney, Andreas Hauptmann, and Samuli Siltanen. Helsinki tomography challenge 2022: description, methods, and results. *Applied Mathematics for Modern Challenges*, 1(1):1–28, 2024. doi: 10.3934/ammc.2024001.
- [27] Hervé Liebgott, Alfonso Rodriguez-Molares, Françoise Cervenansky, Jørgen Arendt Jensen, and Olivier Bernard. Plane-wave imaging challenge in medical ultrasound. In *Proc. IEEE International Ultrasonics Symposium (IUS)*, pages 1–4, 2016. doi: 10.1109/ULTSYM.2016.7728908.
- [28] Saif Khan, Jaeyoung Huh, Manish Bhatt, Md. Murad Hossain, and Arun Bhatt. A deep learning framework for ultrasound image formation and image quality assessment. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, 70(9):972–984, 2023. DeepUS dataset: <https://zenodo.org/doi/10.5281/zenodo.7986407>.
- [29] Hanchen Wang, Tianfan Fu, Yuanqi Du, Wenhao Gao, Kexin Huang, Ziming Liu, Payal Chandak, Shengchao Liu, Peter Van Katwyk, Andreea Deac, et al. Scientific discovery in the age of artificial intelligence. *Nature*, 620:47–60, 2023.

- [30] David J. Brady, Michael E. Gehm, Ronald A. Stack, Daniel L. Marks, David S. Kittle, Dathon R. Golish, E. M. Vera, and Steven D. Feller. Multiscale gigapixel photography. *Nature*, 486(7403):386–389, 2012.
- [31] Yuan Xu, Ti Bai, Hao Yan, Luo Ouyang, Arnold Pompos, Jing Wang, Linghong Zhou, Steve B. Jiang, and Xun Jia. A practical cone-beam CT scatter correction method with optimized Monte Carlo simulations for image-guided radiation therapy. *Physics in Medicine and Biology*, 60(9):3567–3587, 2015. doi: 10.1088/0031-9155/60/9/3567.
- [32] Xiao Liang, Liyuan Chen, Dan Nguyen, Zhiguo Zhou, Xuejun Gu, Ming Yang, Jing Wang, and Steve Jiang. Generating synthesized computed tomography (CT) from cone-beam computed tomography (CBCT) using CycleGAN for adaptive radiation therapy. *Physics in Medicine and Biology*, 64(12):125002, 2019. doi: 10.1088/1361-6560/ab22f9.
- [33] Chenyang Shen, Yesenia Gonzalez, Liyuan Chen, Steve B. Jiang, and Xun Jia. Intelligent parameter tuning in optimization-based iterative CT reconstruction via deep reinforcement learning. *IEEE Transactions on Medical Imaging*, 37(6):1430–1439, 2018. doi: 10.1109/TMI.2018.2823679.
- [34] Eric E. Klein, Joseph Hanley, John Bayouth, Fang-Fang Yin, William Simon, Sean Dresser, Christopher Serago, Francisco Aguirre, Lijun Ma, Bijan Arjomandy, Chihray Liu, Craig Sandin, and Timothy Holmes. Task Group 142 report: Quality assurance of medical accelerators. *Medical Physics*, 36(9):4197–4212, 2009.
- [35] Sasa Mutic, Jatinder R. Palta, Elizabeth K. Butker, Indra J. Das, M. Saiful Huq, Leh-Nien Dick Loo, Bill J. Salter, Cynthia H. McCollough, and Jacob Van Dyk. Quality assurance for computed-tomography simulators and the computed-tomography-simulation process: Report of the AAPM Radiation Therapy Committee Task Group No. 66. *Medical Physics*, 30(10):2762–2792, 2003.
- [36] Inchang Choi, Daniel S. Jeon, Giljoo Nam, Diego Gutierrez, and Min H. Kim. High-quality hyperspectral reconstruction using a spectral prior. *ACM Transactions on Graphics (Proceedings of SIGGRAPH Asia)*, 36(6):218:1–218:13, 2017. doi: 10.1145/3130800.3130810.
- [37] George Cybenko. Approximation by superpositions of a sigmoidal function. *Mathematics of Control, Signals and Systems*, 2(4):303–314, 1989.
- [38] Maziar Raissi, Paris Perdikaris, and George E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.

- [39] Xiaowan Hu, Yuanhao Cai, Jing Lin, Haoqian Wang, Xin Yuan, Yulun Zhang, Radu Timofte, and Luc Van Gool. HDNet: High-resolution dual-domain learning for spectral compressive imaging. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 17542–17551, 2022.
- [40] José M. Bioucas-Dias and Mário A. T. Figueiredo. A new TwIST: Two-step iterative shrinkage/thresholding algorithms for image restoration. *IEEE Transactions on Image Processing*, 16(12):2992–3004, 2007. doi: 10.1109/TIP.2007.909319.
- [41] Xin Yuan. Generalized alternating projection based total variation minimization for compressive sensing. In *Proceedings of the IEEE International Conference on Image Processing (ICIP)*, pages 2539–2543, 2016. doi: 10.1109/ICIP.2016.7532817.
- [42] Tao Huang, Weisheng Dong, Xin Yuan, Jinjian Wu, and Guangming Shi. Deep gaussian scale mixture prior for spectral compressive imaging. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16216–16225, 2021.
- [43] Yuanhao Cai, Jing Lin, Xiaowan Hu, Haoqian Wang, Xin Yuan, Yulun Zhang, Radu Timofte, and Luc Van Gool. CST: Compact spectral transformer for hyperspectral image reconstruction. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2022.
- [44] Yang Liu, Xin Yuan, Jinli Suo, David J. Brady, and Qionghai Dai. Rank minimization for snapshot compressive imaging. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 41(12):2990–3006, 2019.
- [45] Jure Zbontar, Florian Knoll, Anuroop Sriram, Tullie Murrell, Zhengnan Huang, Matthew J. Muckley, Aaron Defazio, Ruben Stern, Patricia Johnson, Mary Bruno, Marc Parente, Krzysztof J. Geras, Joe Katsnelson, Hersh Chandarana, Zizhao Zhang, Michal Drozdal, Adriana Romero, Michael Rabbat, Pascal Vincent, Nafissa Yakubova, James Pinkerton, Duo Wang, Erich Owens, C. Lawrence Zitnick, Michael P. Recht, Daniel K. Sodickson, and Yvonne W. Lui. fastMRI: An open dataset and benchmarks for accelerated MRI. *arXiv preprint arXiv:1811.08839*, 2018.

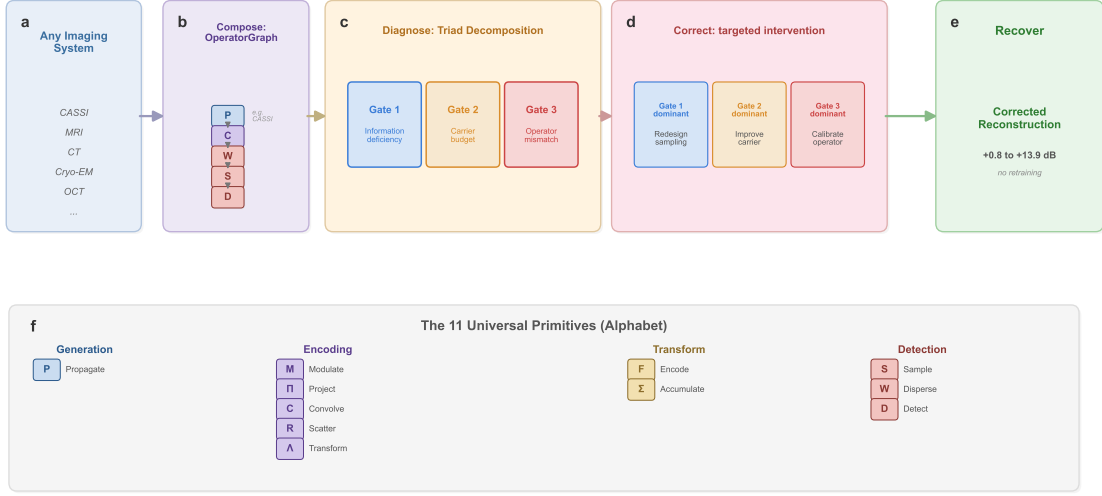


Figure 1: **The universal grammar of computational imaging.** **a**, Any computational imaging system—from coded-aperture cameras to cryo-electron microscopes—serves as input. **b**, The system is composed as a typed directed acyclic graph (DAG) over 11 universal primitives (Figure 2); shown here for CASSI as an example. **c**, The TRIAD DECOMPOSITION diagnoses the DAG through three independent gates: information deficiency (Gate 1), carrier budget (Gate 2), and operator mismatch (Gate 3), each producing a quantitative score (Figure 3). **d**, Based on the dominant gate, a targeted correction is applied: sampling redesign (Gate 1), source/detector improvement (Gate 2), or operator calibration (Gate 3). **e**, The existing solver is re-run with the corrected operator, recovering +0.8 to +13.9 dB reconstruction quality without retraining. **f**, The 11 universal primitives grouped by physical role—generation (Propagate P), encoding (Modulate M , Project Π , Convolve C , Scatter R , Transform A), transform (Encode F , Accumulate Σ), and detection (Sample S , Disperse W , Detect D)—constitute the complete alphabet from which every imaging modality is composed. Together, the 11 primitives (alphabet) and 3 gates (rules) form a universal grammar for designing, diagnosing, and correcting any imaging system.

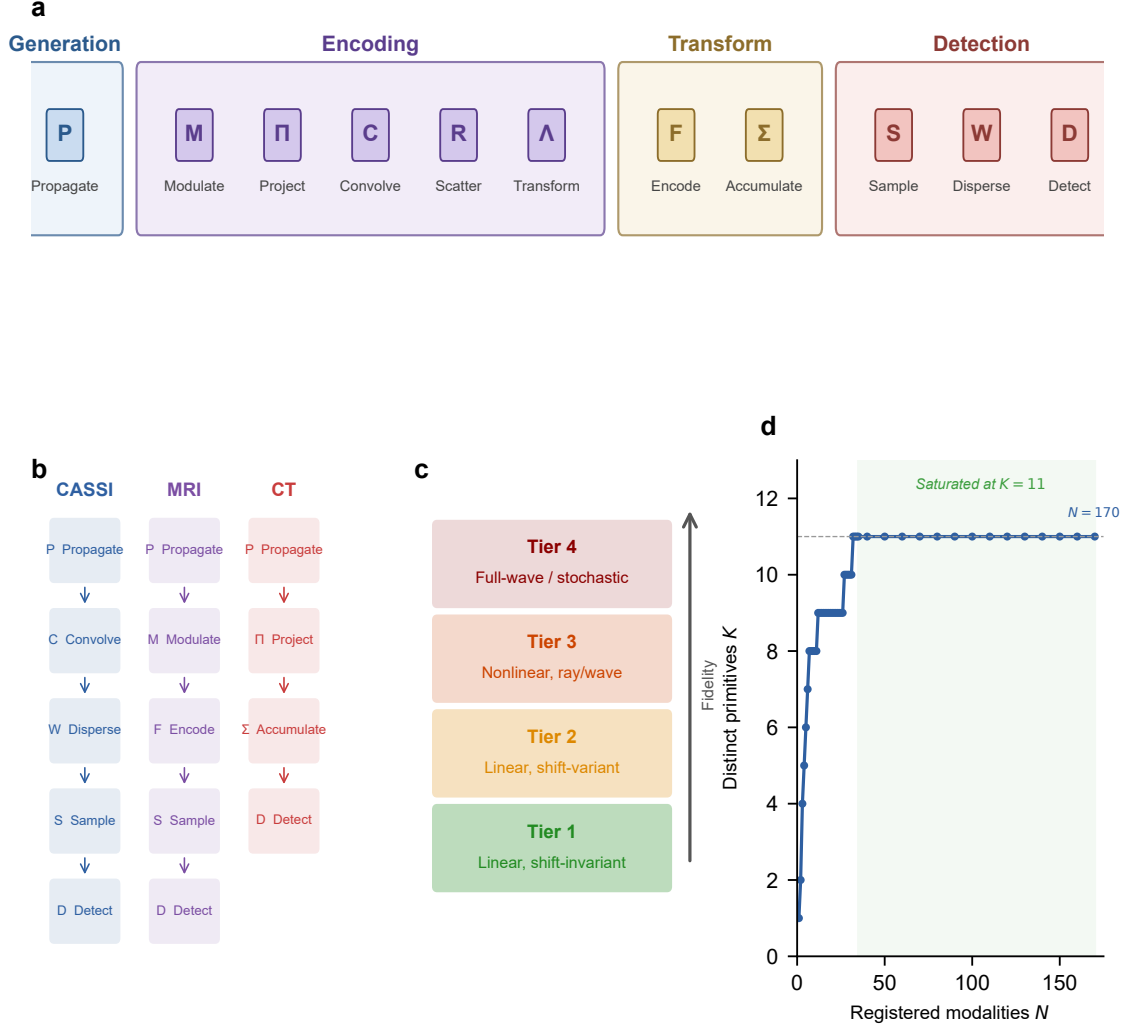


Figure 2: 11 primitives: OperatorGraph IR, Physics Fidelity Ladder, and basis completeness. **a**, The 11 universal primitives grouped by physical role: generation (Propagate P), encoding (Modulate M , Project Π , Convolve C , Scatter R , Transform Λ), transform (Encode F , Accumulate Σ), and detection (Sample S , Disperse W , Detect D). Each primitive carries a typed signature mapping its input to its output (e.g., P : free-space propagation; D : detector response). **b**, Example OPERATORGRAPH DAGs for three modalities: CASSI (photon), MRI (spin), and CT (X-ray photon). Each node wraps a primitive operator; edges define data flow. The symbols in panel b correspond directly to the primitive labels in panel a. **c**, Fidelity levels: forward models range from linear shift-invariant (simplest) through nonlinear and full-wave; all 11 primitives apply uniformly across every level. **d**, Basis-growth saturation: the number of distinct primitives K required to express N registered modalities saturates at $K=11$ for $N \geq 35$; no additional primitive has been needed for the 135 subsequent modalities across 23 physical categories.

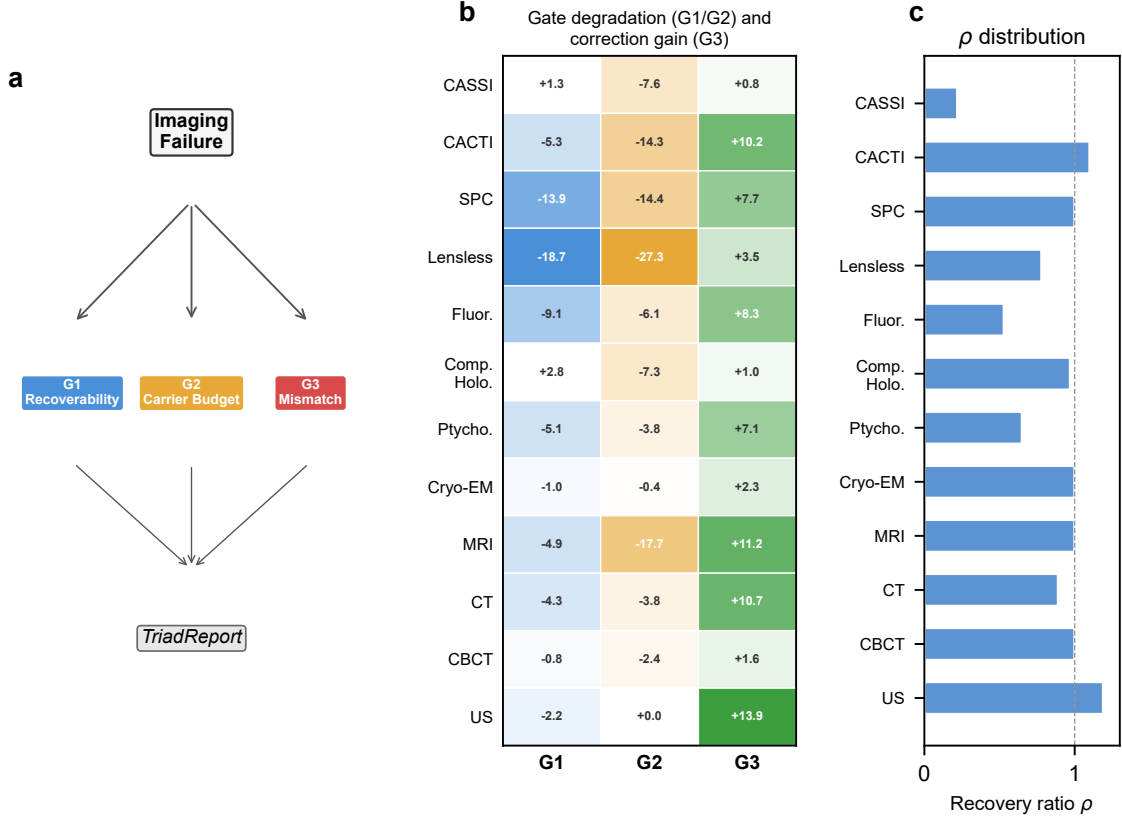


Figure 3: **Triad Decomposition: all three gates validated with real data.** **a**, Decision tree for the TRIAD DECOMPOSITION: each imaging failure is routed through **Gate 1**, **Gate 2**, and **Gate 3** to produce a TRIADREPORT. **b**, Three-gate degradation and correction heatmap. *G1* (blue): Δ PSNR under extreme compression (Table S12; all 12 modalities). *G2* (amber): Δ PSNR under extreme noise (Table S13; all 12 modalities). *G3* (green): correction gain Δ PSNR under standard mismatch conditions (Table S1; all 12 modalities). *G1/G2* degrade at extreme conditions confirming they are real failure modes; *G3* dominates under standard deployment because Gates 1 and 2 were already optimized at design time. **c**, Recovery ratio ρ distribution across all 12 validated modalities, showing that autonomous calibration recovers a substantial fraction of the oracle correction ceiling across all five carrier families.

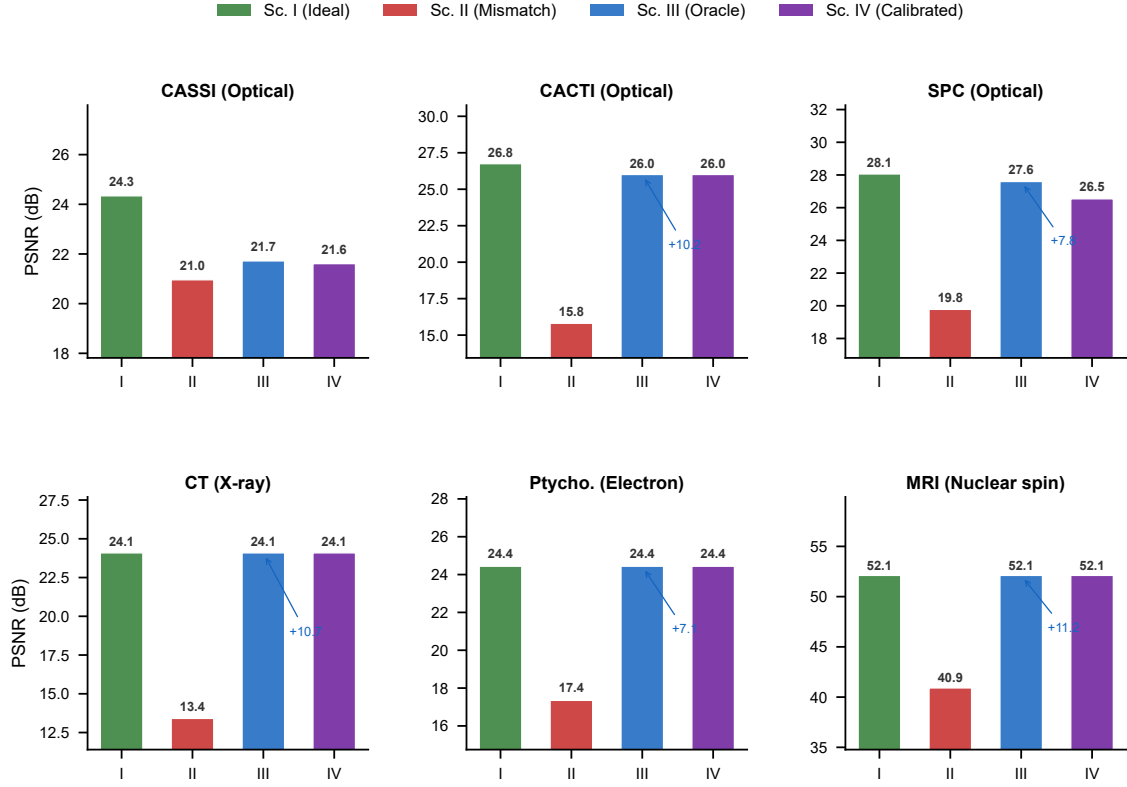


Figure 4: **4-Scenario Protocol across 6 representative modalities.** Each panel shows PSNR (dB) for the four scenarios: Sc. I (ideal operator, green), Sc. II (mismatched operator, red), Sc. III (oracle correction, blue), and Sc. IV (autonomous calibration, purple). Six modalities span four of five carrier families: CASSI, CACTI, and SPC (optical photons); CT (X-ray); ptychography (electron); MRI (nuclear spin). For low-dimensional mismatch (CT, ptychography, MRI), Sc. IV $\approx$ Sc. III $\approx$ Sc. I, confirming near-complete recovery. For multi-parameter mismatch (CASSI), recovery reaches 85%; CACTI and SPC achieve 100% and 86%, respectively (Supplementary Table S9). Sc. I PSNR from Table S3; Sc. II/III from Table S1.

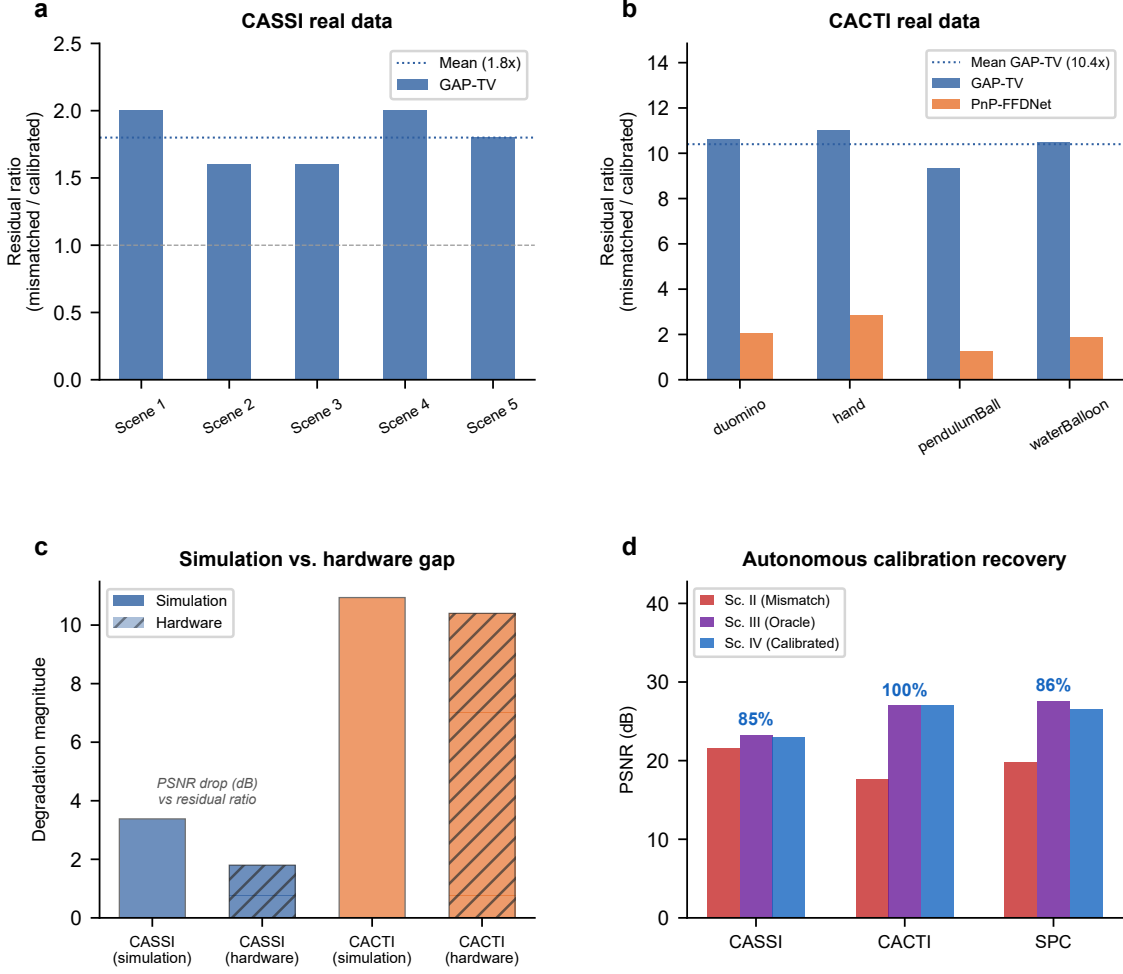


Figure 5: **Hardware validation on real CASSI and CACTI instruments.** **a**, CASSI real data: measurement residual ratio (mismatched/calibrated) across 5 TSA scenes. GAP-TV shows $1.8\times$ mean ratio. **b**, CACTI real data: residual ratio across 4 scenes. GAP-TV shows $10.4\times$ mean ratio; PnP-FFDNet shows $2.0\times$. **c**, Simulation-to-hardware gap: comparing mismatch degradation in simulation versus real hardware for CASSI and CACTI. **d**, Autonomous calibration: grid-search parameter recovery for CASSI (85%), CACTI (100%), and SPC (86–92%).

Extended Data Table 1 | Primitive necessity ablation. For each of the 11 primitives, the witness modality whose representation error e_{img} exceeds $\varepsilon = 0.01$ when that primitive is removed from $\mathcal{B}$. Eight primitives ($P, M, \Pi, F, \Sigma, D, R, \Lambda$) are strictly necessary; the remaining three (C, S, W) are necessary under the stated complexity bounds ($N_{\text{max}} = 20, D_{\text{max}} = 10$). Data from ¹, Proposition 1.

1280

Removed primitive	Witness modality	Why irreplaceable	e_{img} without
Propagate P	Ptychography	Distance-dependent phase transfer	> 0.05
Modulate M	CASSI	Arbitrary element-wise mask	> 0.10
Project Π	CT	Line-integral (Radon) geometry	> 0.15
Encode F	MRI	Non-uniform Fourier sampling	> 0.20
1281 Convolve C	Lensless imaging	Arbitrary shift-invariant PSF kernel	> 0.08
Accumulate Σ	SPC	Dimensional summation (not scaling)	> 0.12
Detect D	All modalities	Carrier-to-measurement conversion	> 0.50
Sample S	MRI	Index-set restriction (k -space)	> 0.10
Disperse W	CASSI	λ -parameterized spatial shift	> 0.06
Scatter R	Compton imaging	Direction change + energy shift	0.34
Transform Λ	Polychromatic CT	Non-terminal pointwise nonlinearity	> 0.05

Extended Data Table 2 | Oracle correction ceiling (Scenario III) across 14 validated configurations (12 modalities, 5 carrier families). Full table with per-scene metrics in Supplementary Table S1; autonomous correction recovery (Scenario IV) in Supplementary Table S9. Per-modality supplementary tables: ultrasound (S14), cryo-EM (S15), CBCT (S16), compressive holography (S17), fluorescence microscopy (S18), SPC (S19), lensless imaging (S20).

1287